# Silicon Nitride Photonic Integrated Devices for Navigation Systems

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# Abstract

Integrating photonic devices into commercial and consumer products has expanded their applications beyond research laboratories. This transition is driven by their compact size, robustness under environmental conditions such as vibrations and temperature variations, and low power consumption. Integrated photonic devices are now expanding their development in navigation technologies by providing accurate, reliable positioning, navigation, and timekeeping (PNT) sensing. Silicon nitride (SiN) photonics has emerged as a leading material due to its low loss, moderate Kerr nonlinearity coefficient, negligible two-photon absorption, and broad optical transparency. These characteristics make SiN an ideal choice for advancing integrated photonic devices, enabling high-accuracy distance and rotational movement measurements for advanced navigation systems. This thesis introduces two SiN-based photonic devices exploiting nonlinear and linear optical phenomena. The thesis discusses the design methodology, experimental validation, and design challenges encountered, along with proposed solutions. A comprehensive study on SiN-based photonic devices for navigation sensor systems is also included.

In the first part of the thesis, we design, simulate, and experimentally characterize a 400 µm SiN microring racetrack resonator with a 30 µm coupling length. The waveguide geometry is optimized for the normal dispersion of -359 ps/nm.km at 1547 nm wavelength. We experimentally characterized the microring resonator, determining a loaded quality factor ($Q_L$) of 77,800 at 1546.19 nm, with an on-chip insertion loss of 0.3 dB. We compare two measured 400 µm radius microring resonators: one in a racetrack configuration and the other a circular ring without a coupling length. The circular ring exhibits a higher insertion loss of 1.5 dB compared to the racetrack microring resonator due to the curvature waveguide in the coupling region. This comparison validates the advantage of the racetrack resonator design for nonlinear optics. In numerical simulations, we demonstrate the resonator's potential to generate six comb lines with a 0.48 nm spacing using two 60 mW pump signals through non-degenerate four-wave mixing (FWM). These comb lines enable precise positioning and timekeeping applications using a LIDAR and optical clock. The second part of the thesis focuses on the development of an efficient double lead zirconate titanate (PZT) actuator-based SiN phase modulator for optical gyroscope applications using piezoelectric and stress-optic effects. By optimizing key geometric parameters, the half-wave voltage ($V_\pi$) is significantly reduced from 65 V to 9.1 V. The optimized design adjusts the distance between two PZT structures from 22.8 µm to 2 µm, reducing the top cladding from 8 µm to 2 µm and increasing the waveguide width from 2.8 µm to 3.7 µm. The maximum tensile stress that mechanically propagates from the PZT structure to the

waveguide structure is $1 \times 10^8$ N/m$^2$ in the core to achieve a $\pi$-shift at low voltage. The optimized design reduces its width by 10 %.

The development of two SiN-integrated photonic devices discussed in this thesis is a key contribution to improving a navigation system's (GNSS) performance through PNT sensors. The integration of an optical frequency comb and an optical gyroscope provides precise distance measurement, timekeeping, and rotational movements of a receiver, potentially enhancing the overall accuracy and reliability of navigation systems.

# Abrégé

L'intégration des dispositifs photoniques dans les produits commerciaux et grand public a élargi leurs applications au-delà des laboratoires de recherche. Cette transition est rendue possible par leur taille compacte, leur robustesse face aux vibrations et aux variations de température, ainsi que leur faible consommation d'énergie. Les dispositifs photoniques intégrés connaissent aujourd'hui un essor dans les technologies de navigation en fournissant des mesures précises et fiables de positionnement, de navigation et de synchronisation (PNT). La photonique à base de nitrure de silicium (SiN) s'impose comme matériau de référence grâce à ses faibles pertes optiques, son coefficient de non-linéarité de Kerr modéré, son absorption biphotonique négligeable et sa large transparence optique. Ces propriétés font du SiN un excellent choix pour le développement de dispositifs photoniques intégrés destinés à des mesures de distance et de mouvements rotationnels de haute précision pour les systèmes de navigation avancés. Cette thèse présente deux dispositifs photoniques en SiN exploitant des phénomènes optiques linéaires et non linéaires. Elle décrit la méthodologie de conception, la validation expérimentale, les défis techniques

rencontrés et les solutions proposées. Une étude approfondie sur les dispositifs photoniques à base de SiN pour les systèmes de capteurs de navigation est également incluse.

Dans la première partie, nous concevons, simulons et caractérisons expérimentalement un résonateur optique à micro-anneau de type racetrack en SiN de 400 µm, avec une longueur de couplage de 30 µm. La géométrie du guide d'onde est optimisée pour une dispersion normale de -359 ps/nm · km à la longueur d'onde de 1547 nm. Le résonateur est caractérisé expérimentalement avec un facteur de qualité chargé ($Q_L$) de 77,800 à 1546,19 nm et une perte d'insertion sur puce de 0,3 dB. Nous comparons deux résonateurs optiques à micro-anneau de 400 µm de rayon: l'un en configuration racetrack, l'autre sous forme annulaire circulaire sans longueur de couplage. Le résonateur circulaire présente une perte d'insertion plus élevée de 1,5 dB en raison de la courbure dans la zone de couplage, ce qui confirme l'avantage de la configuration racetrack pour les applications en optique non linéaire. Des simulations numériques montrent la génération de six lignes de peigne optique espacées de 0,48 nm à l'aide de deux pompes de 60 mW, via un mélange à quatre ondes non dégénéré. Ces lignes de peigne permettent des applications de positionnement et de synchronisation de haute précision utilisant un système LIDAR et une horloge optique.

La seconde partie de la thèse porte sur le développement d'un modulateur de phase SiN basé sur des actionneurs doubles PZT (plomb-zirconate-titanate) pour des applications de gyroscope optique, utilisant les effets piézoélectrique et optique de contrainte. En optimisant certains paramètres géométriques clés, la tension de demi-onde ($V_\pi$) est significativement

réduite de 65 V à 9,1 V. La conception optimisée réduit la distance entre les deux structures PZT de 22,8 µm à 2 µm, diminue l'épaisseur du revêtement supérieur de 8 µm à 2 µm, et augmente la largeur du guide d'onde de 2,8 µm à 3,7 µn. La contrainte de traction maximale transmise mécaniquement de la structure PZT vers le guide d'onde atteint $1 \times 10^8$ N/m$^2$ dans le cœur pour générer un déphasage de $\pi$ à basse tension. Cette conception permet aussi une réduction de la largeur de 10 %.

Le développement des deux dispositifs photoniques intégrés en SiN présenté dans cette thèse constitue une contribution essentielle à l'amélioration des performances des systèmes de navigation (GNSS) via les capteurs PNT. L'intégration d'un peigne de fréquences optiques et d'un gyroscope optique permet des mesures précises de distance, de synchronisation temporelle et de mouvements de rotation d'un récepteur, améliorant ainsi la précision et la fiabilité globales des systèmes de navigation.

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# Contents

**1 Introduction** **1**

1.1 Silicon Nitride . . . . . 3

1.2 Motivation . . . . . 6

1.3 Research Contribution . . . . . 12

1.4 Thesis Organization . . . . . 13

**2 Fundamental Background** **16**

2.1 Optical Frequency Comb . . . . . 16

2.2 Silicon Nitride Microring for OFC Generation . . . . . 19

2.2.1 Microring Resonator Principles . . . . . 23

2.2.2 Nonlinear Optics . . . . . 30

2.2.3 Four-Wave Mixing . . . . . 33

2.2.4 OFC Generation in Microring using Non-degenerate FWM . . . . . 35

2.3 Optical Gyroscope . . . . . 37

2.4 State-of-arts of PZT on SiN Phase Modulator . . . . . 40
2.4.1 Stress-Optic Effect . . . . . 43
2.4.2 Phase Change from Mechanical Stress . . . . . 50

**3 SiN Microring Optical Frequency Comb in Normal Dispersion 52**

3.1 Simulated Design Methodology . . . . . 53
3.2 SiN Design Layout . . . . . 61
3.3 SiN Racetrack Microring Resonator . . . . . 65
3.4 SiN Circular Microring Resonator . . . . . 69
3.5 Optical Frequency Comb Simulation . . . . . 73
3.6 Summary . . . . . 76

**4 Actuator-based Silicon Nitride Phase Modulator 78**

4.1 Design Specifications . . . . . 79
4.2 Design of the Single PZT Actuator-based SiN Phase Modulator . . . . . 80
4.3 Performance Analysis . . . . . 85
4.3.1 PZT Separation Optimization . . . . . 88
4.3.2 Top Cladding Thickness Optimization . . . . . 90
4.3.3 Core Waveguide Width Optimization . . . . . 91
4.4 Summary of the Optimization Results . . . . . 93

**5 Conclusion 95**

5.1 Thesis Summary . . . . . 95

5.2 Future Works . . . . . 97

# List of Figures

1.1 Illustration of various approaches to the positioning, navigation, and timing (PNT) in three key sensors: LIDAR for positioning, an optical gyroscope for navigation, and an optical clock for timekeeping. LIDAR and the optical clock are developed using an optical frequency comb, while the optical gyroscope operates independently as a standalone navigation sensor. Together, these sensors integrate into PNT solution to enhance the accuracy of the global navigation satellite system (GNSS). . . . . 8

2.1 Mode-locked laser-based optical frequency comb representation. (a) In the time domain, a pulse train from a mode-locked laser shows slippage $\Delta\phi_{\mathrm{CEO}}$ between the carrier phase (thick red line) and the pulse envelope (solid blue line) from one pulse to the next. (b) In the frequency domain, this phase slippage $\Delta\phi_{\mathrm{CEO}}$ leads to the carrier-envelope offset frequency from the Fourier transform between the time and frequency domain. The frequency repetition rate ($f_{\mathrm{rep}}$) represents the spacing between comb lines [taken from Schilt et al. [58]]. © 2015 by the authors; licensed under the terms and conditions of the Creative Commons Attribution license (http://creativecommons.org/licenses/by/4.0/) . . . . . 18

2.2 (a) Waveguide dispersion for varying SiN core thickness, blue refers to normal dispersion ($D < 0$) and red refers to ($D > 0$); (b) SiN waveguide bending loss versus bending radius of the TE0 optical mode profile for different SiN core thicknesses (taken from [28] with copyright permission from the Photonics Research Journal). . . . . 20

2.3 (a) Single-bus microring resonator; (b) basic design of the ring directional coupler ($t$: transmission coupling, $k$: field coupling)(c) normalized optical transmission response (ER: extinction ratio, FSR: free-spectral range, FWHM: full wave half maximum) . . . . . 24

2.4 Conceptual illustration of the FWM nonliear process using the energy level band diagram and the corresponding spectral of the wavelength conversion leading to an idler ($\omega_i$) and signal ($\omega_s$): (a) de-generate FWM when the optical pump signals have the same frequency ($\omega_p$); (b) non-degenerate FWM when their frequencies are different ($\omega_{p1}$ and $\omega_{p1}$). . . . . 34

2.5 Illustration of OFC generation in a microring resonator through non-degenerate FWM.. . . . . 36

2.6 Illustration of the Sagnac effect in a ring waveguide. When stationary, two light beams injected at point M follow identical optical paths. In a clockwise rotation, the clockwise beam (red dashed line) experiences a longer effective optical path (displacing point M to M') compared to the counterclockwise (CCW) beam (blue dashed line) displaced to M" [104]. (This open-access article is distributed under the terms of the Creative Commons Attribution license (http://creativecommons.org/licenses/by/4.0/) . . . . . 39

2.7 Cross-section schematic illustrating the stress (red straight lines between the two metal electrodes) in an optical modulator using a PZT actuator above a SiN waveguide surrounded by silicon dioxide $SiO_2$ in a silicon (Si) photonic integrated technology platform. The stress comes from the applied voltage between the two metal electrodes causing compressive (blue dashed lines) and tensile stresses (red dashed lines) in SiN waveguide geometry leading to a change in the effective index. . . . . 42

2.8 The PZT actuator operates based on the inverse piezoelectric effect. This causes displacement (illustrated by yellow arrows) in the PZT material, which is sandwiched between two metal layers. The displacement occurs in the PZT layer due to the presence of an applied voltage to the metal layers. . . . . 44

3.1 (a) Cross-section of SiN technology platform b) Schematic of the 400 µm SiN microring resonator including coupling length ($L_c$) and coupling gap ($g$), and the radius ($R$). . . . . 55

3.2 (a) FDE simulated TE0 mode normalized intensity profile in the SiN waveguide; (b) plot of the effective mode index $n_{eff}$ as a function of wavelength for a 400 nm thick and 2 µm wide cross-sectional SiN waveguide. 56

3.3 Dispersion coefficients of a 400 nm thick SiN waveguide with a bend radius of 400 µm as a function of wavelength. (a) Comparison of dispersion profiles for waveguide widths ranging from 0.5 µm to 2.5 µm. (b) Zoomed-in view of the dispersion for the 2 µm wide waveguide. . . . . 57

3.4 Normalized transmission (top row) and zoomed-in around 1546.56 nm (bottom row) of the simulated SiN 400 µm microring racetrack resonator. Results with a) a propagation loss of 10.5 dB/cm for the ring waveguide and 1.75 dB/cm for the straight waveguide and (b) with a propagation loss of 3.2 dB/cm for the ring and 2.2 dB/cm for the straight waveguide. . . . . 59

3.5 (a) Chip layout with various SiN microring resonators and test structures. The red dotted rectangle indicates the measured 400 µm SiN racetrack microring, while the green dotted rectangle highlights the measured 400 µm SiN circular microring; the dark blue dotted rectangles indicate the other measured three microrings with radii of 50 µm, 83 µm and 100 µm; (b) zoomed-in view of the red rectangle labeled microring resonator with vertical grating couplers. . . . 62

3.6 a) Schematic of the experimental test setup used to measure the fabricated SiN circular and racetrack microring (photo image shows the racetrack ring from the fabricated die) (DUT: device under test, CW: continuous wave). b) Photography of the experimental laboratory setup with labeled key testbed components. . . . . 65

3.7 Measured optical transmission of the loopback structure with two SiN vertical grating couplers. The optical input power is 0 dBm. . . . . 67

3.8 (a) Measured transmission response of the 400 µm SiN racetrack resonator, (b) zoomed-in view at 1546.19 nm resonating wavelength. . . . . 69

3.9 (a) Normalized transmission response obtained through simulation with a propagation loss of 3.2 dB/cm for the ring and 2.2 dB/cm for the bus waveguide, (b) zoomed-in view around 1546.2 nm wavelength . . . . . 70

3.10 a) Measured normalized transmission spectrum of the 400 µm circular microring with b) a zoomed-in view of resonance at 1545.59 nm wavelength. 71

3.11 Simulated results of the OFC spectrum in a 400 µm radius SiN microring racetrack resonator for pump wavelengths at 1546.56 nm and 1547.02 nm. (a) Two equal-power pumps with input power of 27 mW (14.3 dBm). (b) Two equal-power pumps with input power of 60 mW (17.8 dBm). . . . . 74

4.1 Cross-sectional view of the double PZT on a SiN waveguide for an actuator-based stress-optic phase modulator. . . . . 80

4.2 Stress tensor obtained through simulation in the xy-plane for a single PZT actuator with an applied voltage of 40 V on the bottom electrode and 0 V on the top electrode. The blue shading represents compressive stress (negative stress) which reaches the SiN waveguide, and the red-to-yellow shading indicates tensile stress or decompression (positive stress). . . . . 82

4.3 (a) Y-stress tensor distribution in the SiN waveguide as a function of the applied voltage across the full width of the PZT structure.(b) effective refractive index change of the optical fundamental mode in the SiN waveguide as a function of the applied voltage on the PZT structure. (c) corresponding phase shift as a function of the applied voltage on the PZT actuator and the y-direction compression stress. . . . . 83

4.4 Resulting y-stress tensor through simulation stress induced from the PZT structure at an applied voltage of 40 V. The blue shading represents compressive stress (negative stress) and red shading represents tensile stress (positive stress). . . . . 86

4.5 (a) Y-stress tensor in the SiN waveguide as a function of the applied voltage on the double PZT structures in Design 1. (b) effective index change as a function of applied voltage on PZT actuators (single mode profile shown in the inset) (c) corresponding phase in the waveguide over different applied voltage. . . . . 87

4.6 Simulated y-stress tensor induced in the SiN waveguide with a 10 µm distance between the two PZT structures at an applied voltage of 40 V on the bottom electrode and 0 V on the top electrode. The blue shade represents negative (compressed) stress and the redish-yellow shade represents positive (tensile) stress. . . . . 88

4.7 (a) Plot of y-stress tensor in the SiN waveguide and (b) Plot of effective index change as functions of applied voltage on the double PZT structures of Design 2. (c) Phase shift change as a function of applied voltages of two PZT separations, 22.8 µm and 10 µm distance. . . . . 89

4.8 Simulated y-stress tensor on the SiN waveguide with a 3 µm top cladding and a 3 µm separation between the PZT structures in Design 3. The blue shade represents negative stress, and the red shade represents positive stress. . . . 90

4.9 (a) Y-stress tensor in the SiN waveguide and (b) effective index change as functions of applied voltage on the double PZT structures of Design 3. (c) corresponding phase shift as a function of applied voltage for Design 3. . . . 91

4.10 Simulated y-component of the stress tensor on the SiN waveguide with a 2 µm cladding and a 2 µm distance between the PZT structures in Design 4. The blue shade represents negative stress and the red shade represents positive stress. . . . . 92

4.11 (a) Y-stress tensor in the SiN waveguide and (b)effective index change as functions of applied voltage on the double PZT structures of Design 4. (c) corresponding phase shift as a function of applied voltage for Design 4. Design 4 consists of 2 µm thick $SiO_2$ top cladding, 2 µm distance between two PZT actuators, and 3.7 µm width of SiN core waveguide. . . . . 92

5.1 Proposed an experiment testbed for OFC generation in the SiN microring racetrack resonator. EDFA: Erbium-doped fiber amplifier (Pritel, LNHPFA-33-IO-NMA), PC: polarization controller, OSA: optical spectrum analyzer. . 98

5.2 The cross-sectional waveguide with the PZT actuation featuring an asymmetric double stripe (ADS) TriPleX™ silicon nitride waveguide. . . . . 99

# List of Tables

3.1 Summary of Simulated Racetrack Microring Performance . . . . . 61

3.2 Design Geometries of Measured Circular and Racetrack Microrings . . . . . 63

3.3 Experimental performance comparison of the SiN racetack and circulator microring resonators. . . . . 71

4.1 Performance comparison between PZT actuator-based SiN modulators . . . 94

# List of Acronyms

| | |
|---|---|
| **ADS** | Asymmetric Double Stripe. |
| **CMOS** | Complementary Metal–Oxide–Semiconductor. |
| **CW** | Continuous Wave. |
| **DC** | Directional Coupler (also Direct Current). |
| **EDFA** | Erbium-Doped Fiber Amplifier. |
| **ER** | Extinction Ratio. |
| **FSR** | Free-Spectral Range. |
| **FWHM** | Full Width Half Maximum. |
| **FWM** | Four-Wave Mixing. |
| **GC** | Grating Coupler. |
| **GNSS** | Global Navigation Satellite System. |
| **GPS** | Global Positioning System. |
| **GVD** | Group Velocity Dispersion. |
| **INS** | Inertial Navigation System. |

| | |
|---|---|
| **LIDAR** | Light Detection and Ranging. |
| **MEMS** | Micro-Electro-Mechanical Systems. |
| **MZI** | Mach Zehnder Interferometer. |
| **NIST** | National Institute of Standards and Technology. |
| **OFC** | Optical Frequency Comb. |
| **OSA** | Optical Spectrum Analyzer. |
| **PNT** | Positioning, Navigation, and Timekeeping. |
| **Pt** | Platinum. |
| **PZT** | Lead Zirconate Titanate. |
| **Q-factor** | Quality factor. |
| **SiN** | Silicon Nitride. |
| **SiO2** | Silicon Dioxide. |
| **SiPh** | Silicon Photonics. |
| **SOI** | Silicon-on-Insulator. |
| **TE** | Transverse Electric. |
| **TeO2** | Tellurium Oxide. |
| **TiW** | Titanium-Tungsten. |
| **TPA** | Two-Photon Absorption. |

# Chapter 1

# Introduction

The increasing demand for navigation technology in autonomous vehicles, drones, and aerospace systems is being driven by the rapid expansion of autonomous and space industries, as well as growing user adoption. As these technologies become more integrated into daily life, the need for advanced, reliable, and precise navigation solutions has become a critical focus for researchers, developers, and manufacturers [1]. The challenge is to create navigation systems that can operate effectively in various environments, ensuring accuracy and stability in real-time applications. Over the past decade, the space industry has witnessed a significant shift in satellite technology, with the development of small satellites revolutionizing the sector. Instead of relying on a single, large satellite, modern space missions often deploy constellations of smaller satellites that work together to perform complex tasks [2]. This shift has introduced new challenges in terms of stabilization, precise

control, and coordination of these small satellites, which are essential for applications in autonomous vehicles, telecommunications, and space industries. The traditional methods of satellite navigation and control are insufficient to overcome the challenges, necessitating the development of more advanced technologies. To address these challenges, integrated photonic circuits (PICs) have emerged as a promising solution for developing precise, compact, and energy-efficient navigation systems. PICs offer numerous advantages over traditional electronic systems, including high bandwidth, lower power consumption, and reduced costs [3]. Their compact and straightforward design allows for the integration of multiple functions into a single chip, significantly reducing the size and weight of the navigation systems [4]. In addition to their performance benefits, PICs provide immunity against electromagnetic interference (EMI), which is crucial in environments where electronic interferences can disrupt signals. They also exhibit excellent mechanical properties, such as flexibility, compactness, and high resistance to vibrations and shocks. Moreover, PICs offer significant immunity to radiation, making them ideal for autonomous and space applications where exposure to high levels of radiation is a concern [4]. The applications of integrated photonics extend beyond navigation systems [5]. They are also being increasingly utilized in fields such as high-speed communications, data centers [6], medical sectors, microwave photonics [7], nonlinear photonics [8], and sensing [9].

## 1.1 Silicon Nitride

Silicon nitride (SiN) photonics is compatible with complementary metal oxide semiconductor (CMOS) processes. It exhibits low propagation loss, broadband transparency, and the ability to handle high optical intensities, up to several watts [10, 11]. SiN leverages a refractive index (1.98 to 2) at the telecommunication wavelength of 1550 nm, enabling the miniaturization of optical devices to the micro/nanoscale. The low index contrast with the surrounding silicon dioxide cladding material (refractive index of 1.45) facilitates a low propagation loss of less than 0.1 dB/cm at approximately 1550 nm [12]. The reduced index contrast of SiN waveguides minimizes sidewall-roughness-induced scattering loss [13]. Significant progress has been achieved with the advancement of SiN photonics in passive integration (e.g., waveguides, resonators, couplers), nonlinear photonics, and hybrid and heterogeneous photonics integration (e.g., modulators, amplifiers, phase shifters, photodetectors) over the past few decades. Basic SiN waveguide structures with $SiO_2$ cladding were first demonstrated in the 1970s, and integrated SiN photonic devices emerged in the 1990s [14, 15]. One of the first single waveguide stripe SiN waveguides developed had a propagation loss of 1-2 dB/cm [16]. In 2009, DARPA's Integrated Photonics for High-Performance Operational Detectors (iPHOD) program aimed to develop on-chip SiN device technologies by reducing device losses and enhancing performance. The goal was to develop compact, cost-effective, efficient, and high-power handling waveguides to improve the performance of applications

such as RF microwave photonic links, optical gyroscopes, and packet storage [17]. RF microwave photonic links transmit RF signals over optical fibers. SiN waveguides of low loss and high-power handling capabilities were leveraged to enhance signal integrity and bandwidth. This enhancement aimed to improve the performance of communication systems. SiN's stability and low loss are also crucial for optical gyroscopes [18]. These properties improved the precision and reliability of rotational movement sensing for navigation in autonomous vehicles and aerospace systems. The development of SiN-based waveguides also aimed to enable packet storage applications by providing efficient and compact solutions for optical buffering and delay lines, crucial for managing data flow in optical networks [19].

SiN material also enabled prominent development in nonlinear integrated devices from 2009. In the early stages, nonlinear integrated photonic devices were fabricated using silica [20] and silicon [21]. However, silicon waveguides suffer from relatively higher propagation loss (1 to 3 dB/cm) due to scattering losses associated with sidewall roughness [22]. While silicon exhibits a significant Kerr nonlinearity coefficient, its effectiveness is considerably reduced in the telecom band due to high levels of two-photon absorption (TPA). TPA limits nonlinear responses and introduces additional waveguide losses at high power levels [23].In contrast, SiN material is highly advantageous for developing nonlinear photonic devices due to its large bandgap (5.6 eV), resulting in negligible TPA. This makes SiN an efficient material for third-order nonlinearity, such as

four-wave mixing (FWM) over a broad bandwidth [24]. Consequently, SiN photonic wires have demonstrated frequency comb and supercontinuum generation [25, 26]. SiN-based integrated photonics devices continue to advance in developing ultra-low loss waveguides for applications ranging from positioning, navigation, and timekeeping (PNT) sensors for navigation systems [27], high-speed data communication to the biomedical sectors [28].

While the contribution reported in this thesis focuses on developing SiN photonic devices for navigation systems, several key challenges, particularly in achieving low optical loss and efficient nonlinear optical processes, are common for optical communication and biosensing applications. A coherent and broadband OFC using the FWM effect is required for high-precision navigation, optical communication, and biosensing. However, one of the significant difficulties is achieving efficient FWM under normal dispersion conditions in the SiN platforms. The efficient FWM in SiN platforms requires anomalous dispersion, which is achieved by dispersion engineering in thick SiN waveguides. These thick SiN layers are prone to stress-induced cracking, posing a significant fabrication challenge that limits the yield and device reliability across all application areas [25, 29]. Another challenge is achieving ultra-low optical loss in SiN platform. While low loss is desirable across all photonic applications, the low loss requirement is most stringent in gyroscope-based navigation (less than 0.1 dB/cm), where loss directly impacts resonator Q-factors and sensitivity [111]. In contrast, communication and biosensing systems can generally tolerate higher losses (1 to 3 dB/cm), though reducing the loss in the SiN platform still enhances

system performance and scalability [30].

## 1.2 Motivation

High precision navigation technology plays a wide range of scientific and technological applications, including rescue operations [31], military and defense systems [32], autonomous vehicles [33], drones, and climate monitoring. The challenge lies in developing navigation systems that ensure accuracy, stability, and reliability in real-time scenarios. A key technology in this domain is the Global Navigation Satellite System (GNSS), which uses a constellation of earth-orbiting satellites to provide precise positioning data. GNSS determines position by transmitting signals from satellites to ground-based receivers, which calculate their location by measuring the time delay between the signal's transmission and its reception. GNSS offers advantages such as high accuracy, low cost, and no accumulation of positioning errors with time [34]. Despite these advantages, GNSS suffers from limitations including a requirement for line-of-sight (LoS) between the receiver and satellite, signal disruption in dynamic environments, and a slower data update rate [32]. To address these challenges, PNT solutions have evolved, incorporating advanced optical technologies for significant improvements in GNSS architecture. Among several developed optical technologies, integrated photonic devices stand out for their compactness, low power consumption, and high optical bandwidth, making them ideal for PNT solutions.

PNT solutions encompass sensors, including light detection and ranging (LIDAR), optical

gyroscopes, and optical clocks [35,36] as in Fig. 1.1. LIDAR is a remote sensor that precisely measures distances between the sensor and an object by emitting laser pulses and analyzing the time it takes for the light to reflect off objects. It is used for object tracking and solving positioning problems [37]. An optical gyroscope is a sensor that measures rotational movements by detecting phase differences between counter-propagating light waves based on Sagnac effect [38]. Optical gyroscope uses as a navigation sensor which measures precise rotational movement of the object [39]. An optical clock is a highly precise timekeeping sensor that uses the oscillations of light (optical frequencies) to measure time [40]. Optical clock requires high frequencies for atomic transition between energy levels [41].An optical clock accurately measures the precise time at which a moving object reaches its destination. It also determines the travel time of signals transmitted from a satellite to its receiver in PNT systems. As illustrated in Fig. 1.1, key component in developing LIDAR and optical clock sensors is the photonic integrated optical frequency comb (OFC) laser, which generates an equally spaced spectrum of wavelengths across a broad optical bandwidth [42]. On-chip OFC provides low power consumption and high resolution [43]. OFC enables the precise distance measurement with high resolutions less than 100 fs as it generates pulses in the femtosecond range, much shorter than typical nanosecond pulses used in conventional LIDAR [44]. Recent research shows an OFC-based LIDAR using interferometric technique measuring phase differences between the emitted and reflected light to precise distance determination up to 30 km [45]. In optical clock applications, the OFC provides a sufficiently stable reference

for comparing atomic frequencies, leading to accurate timekeeping sensor [46].

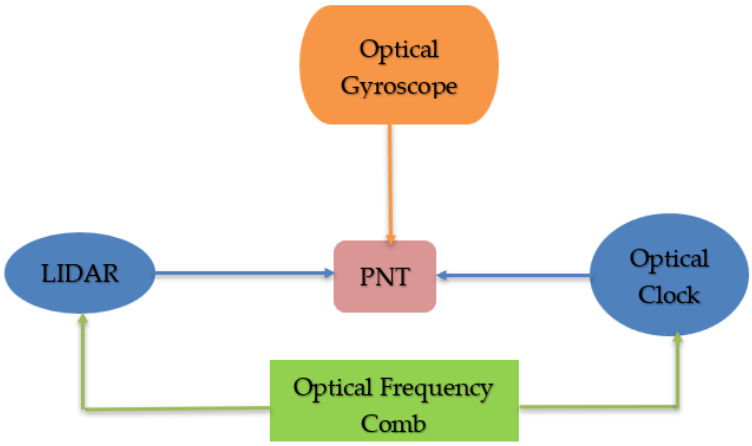



**Figure 1.1:** Illustration of various approaches to the positioning, navigation, and timing (PNT) in three key sensors: LIDAR for positioning, an optical gyroscope for navigation, and an optical clock for timekeeping. LIDAR and the optical clock are developed using an optical frequency comb, while the optical gyroscope operates independently as a standalone navigation sensor. Together, these sensors integrate into PNT solution to enhance the accuracy of the global navigation satellite system (GNSS).

Beyond the possible PNT applications such as LIDAR and optical clocks, optical gyroscopes are another important category (Fig. 1.1) that can precisely measures the rotational movement of an object. Optical gyroscopes were initially developed using high-quality optical fibers. Although extending the fiber length in the Sagnac cavity still allows for relatively compact devices, the bending loss becomes significant [47, 48]. To address these limitations, resonant integrated passive optical gyroscopes emerged as a viable alternative to traditional fiber optic gyroscopes. These resonant gyroscopes offer several advantages, including lower loss, more compact size, and reduced cost [49].

However, the resonant gyroscope requires a high-performance modulator to improve precise rotational sensitivity by controlling optical signals within the gyroscope and detecting very small phase differences, even when the gyroscope's rotation is close to zero.

This thesis explores and investigates two SiN photonic devices for navigation system applications through PNT solutions (Fig. 1.1). The first part of the thesis is on a SiN microring resonator design optimization using normal waveguide dispersion to generate an optical frequency comb (OFC) through non-degenerate FWM. By simulating and optimizing the geometry of the core waveguide and the radius of the resonator, the design aims to achieve the desired normal dispersion. The fabrication and experimental characterization are conducted to obtain the key parameters, demonstrating the potential of OFC generation through numerical simulations. The novelty of this work lies in achieving a frequency comb in the normal dispersion regime through FWM. In the literature, FWM typically requires either zero anomalous or zero dispersion to operate effectively. The efficiency of FWM in normal dispersion regimes is reduced due to the combined effects of phase mismatch and spectral line broadening. In normal dispersion, the negative waveguide dispersion causes different wavelength components to travel at different speeds, leading to a phase mismatch. This phase mismatch reduces the coherent interaction between pump waves required for an efficient FWM. Additionally, dispersion broadens the spectral lines of the interacting pump waves, reducing their spectral overlap and temporal coherence, further reducing the FWM conversion efficiency. Despite the

challenges of achieving FWM in normal dispersion, normal dispersion is achieved from a thin SiN film which is less prone to cracking in its fabrication. Therefore, it is interesting to investigate OFC generation from FWM in the normal dispersion regime. As discussed earlier, the efficiency of FWM in normal dispersion regimes is inherently limited, resulting in the generation of a relatively small number of equally spaced comb lines. These short combs are well-suited for applications such as short-range LIDAR, where high precision in distance measurement is critical. A short-range LIDAR example is the Short Comb Atmospheric Lidar Experiment (SCALE) led by CNES (Centre National d'Études Spatiales), which employs dual-comb spectroscopy technique for precise ranging up to 50 km distance and operates near 1550 nm wavelength. An experimental study reported in the literature demonstrated that the SCALE (Short Comb Atmospheric Lidar Experiment) system employed between 3 to 13 optical frequency comb lines, validating its performance across two distinct scenarios [50]. In the first scenario, 9 comb lines were used, with a minimum output power of -18.88 dBm; in the second, 13 comb lines were included, though the weakest lines reached down to -32.3 dBm. Comb lines below this output power threshold are considered too weak to be detected by photodetectors due to insufficient signal strength [50]. In this thesis, a 400 µm SiN microring resonator is proposed to generate 15 comb lines in normal dispersion using the FWM technique, including two strong pump sources at 17.8 dBm. Among 15 comb lines, only 6 comb lines exhibit output powers above -22 dBm, which are sufficient for detection and meeting the requirements of

SCALE implementations. The remaining comb lines are below -40 dBm and are therefore excluded from use due to insufficient optical power. Therefore, only six usable comb lines are targeted for implementation in a short-range LIDAR system, directly supporting PNT applications. These comb lines enable accurate ranging in integrated photonic navigation platforms, offering a compact, low-power, and scalable solution for optical PNT systems.

The second design reported in this thesis is on the development of a stress-optic phase modulator through the course of a Mitacs-funded internship with Montreal-based startup One Silicon Chip Photonics (OSCP). The stress-optic phase modulator consists of a customized double PZT actuator on a SiN waveguide to enhance the sensitivity and performance of an optical gyroscope as a navigation sensor in a PNT solution (Fig. 1.1). PZT actuators convert electrical energy into mechanical stress, deforming the PZT structure when voltage is applied across its electrodes. This stress propagates to the SiN waveguide, altering its shape and changing the effective refractive index. As a result, the phase of the optical signal in the waveguide is modulated. This configuration is used in a PZT-based SiN stress-optic phase modulator, which modulates one or both injected light beams to shift the baseline in OSCP's gyroscope, and enhances the rotational sensitivity when the gyroscope operates at low rotation rate. The details regarding the integration of the designed PZT actuator-based SiN phase modulator within OSCP's optical gyroscope are confidential and cannot be disclosed. The stress-optic phase modulator design optimization approach looks to vary the geometries of the PZT-SiN-based waveguide

structure, with each configuration characterized by effective index change, applied voltage, and electrode lengths through simulation. The key to OSCP developers in our collaboration was to optimize the design to obtain maximum phase shift at relatively low applied voltage. A recent publication reports that a maximum phase shift of $\pi$ (V$\pi$) in piezoelectric MEMS gyroscopes requires up to 66 V due to the combined influence of piezoelectric and electrostatic effects [51]. However, in this thesis, we focus solely on piezoelectric-based phase modulation using double PZT actuators integrated with a SiN waveguide. The aim is to reduce the $V_\pi$ and improve the sensitivity and energy efficiency of the SiN-based optical gyroscope, which is a key component in an optical PNT system. The proposed design achieves a $V_\pi$ of 9.1 V, significantly lower than previously reported values, enabling practical deployment in integrated photonic circuits. In the context of PNT applications, where precise phase control and low power consumption are critical, an ideal V$\pi$ target lies in the 1-10 V range. Achieving such low driving voltage not only improves the rotational sensitivity of the optical gyroscope but also enables energy-efficient implementation of next-generation PNT systems.

## 1.3 Research Contribution

In July 2024, I presented the research work related to the SiN microring resonator-based OFC generation at the Advanced Photonics Congress held in Quebec city:

[142] **R. Nazneen** and O. Liboiron-Ladouceur, "Silicon Nitride Microring Resonator

for Optical Frequency Comb Generation in Normal Dispersion", *Optica Advanced Photonics Congress*, paper ITu3B.2, Quebec City, Canada, 28 July - 01 August 2024. Further, a detailed report entitled "Design Report on Double PZT Actuator-Based Silicon Nitride Phase Modulator" on the design specifications of various PZT-based SiN phase modulators, including their simulated results and key performance metrics, was presented and submitted to the OSCP collaborators. A US patent application was submitted in January 2025.

[156] K. Zandi, L. R. Williston, R. Tewari, M. Fasihanifard, O. Liboiron-Ladouceur and **R. Nazneen**, 'Integrated Phase Modulator', US Provisional Patent Application No. 63/747,217, January 20, 2025.

## 1.4 Thesis Organization

The thesis encompasses the modeling and performance analysis of essentially two silicon nitride-based integrated devices for navigation systems through PNT solutions. The thesis is structured into five chapters, with the current chapter as an introduction. This introductory chapter presented an overview of integrated photonic devices and silicon nitride, highlighting their respective features and limitations towards PNT solutions. Additionally, it outlined the motivation behind the research projects based on recent advancements in the field, and my research contributions made.

Chapter 2 overviews optical frequency combs (OFC) and their application in navigation

systems. Also, it provides a study on recent advancements in SiN microring resonators for OFC generation, along with a comprehensive overview of the fundamental principles of microring resonators, including a description of their key parameters. The chapter also covers essential concepts in nonlinear optics, including FWM, dispersion, and losses for OFC generation. Furthermore, it provides a recent advancement study on an optical gyroscope for rotational movement sensing for navigation system applications. The state-of-the-art developments related to PZT-based SiN phase modulators improve the performance of a gyroscope. The chapter also includes relevant concepts such as the stress-optic effect, piezoelectric effect, and the relationship between phase change, refractive index, applied voltage, and electrode length.

Chapter 3 outlines the design specifications and simulation methodologies for silicon nitride (SiN) microring resonators. The design of the microring begins with the selection of suitable materials, parameters, and components. The waveguide geometry (width and radius) is first optimized through simulation to achieve the desired dispersion characteristics. Next, the microring geometry (coupling length and gap) is optimized to tailor the transmission response and the Q factor. Once the transmission characteristics are verified, replicas of the optimized design with varying geometries are included in the fabrication layout for comparative analysis. An experimental test plan is introduced to validate the microring performance, and the measured results are analyzed to confirm the design objectives. Finally, the chapter presents a simulated optical frequency comb (OFC)

spectrum based on the characterized parameters, along with the corresponding simulation results.

Chapter 4 details the design specifications and simulation methodology for the SiN phase modulator based on PZT actuators. The methodology begins with the selection of parameters, components, and materials that are sourced from OSCP and reference literature. A single PZT actuator-based modulator is first designed and analyzed to evaluate its phase shift response for driving voltage. Building on this, an efficient double PZT actuator-based SiN phase modulator is developed in collaboration with the OSCP team. Design optimization focused on key geometric parameters, including the spacing between the PZT actuators, the thickness of the cladding at the top $SiO_2$ and the width of the waveguide core. After each design iteration, the stress distribution and effective index changes are analyzed under applied voltages to determine the resulting phase shift. The aim is to identify the optimal design structure for enhancing the performance of an optical gyroscope.

Chapter 5 summarizes the development made through this work related to SiN photonic devices. Based on the results obtained, the chapter outlines possible future works, including a proposed experimental setup for OFC generation, a detailed fabrication process, and experimental methodology for validating a PZT-actuated SiN modulator intended for use in navigation system applications.

# Chapter 2

# Fundamental Background

## 2.1 Optical Frequency Comb

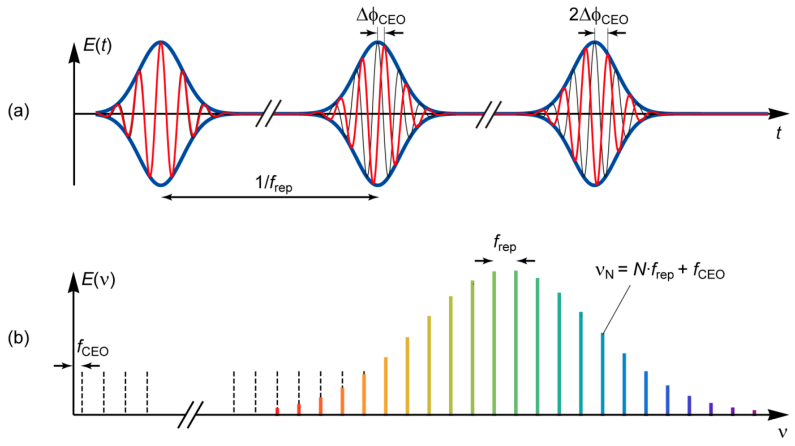
An optical frequency comb (OFC) is a light source whose optical spectrum consists of discrete and equally spaced spectral lines. OFC exhibits an even interval of multiple optical frequencies in the frequency domain. The spacing between adjacent comb lines is constant. OFC is a photonic technology that revolutionized the scientific and industrial fields such as high-capacity optical communication [52], biosensing, meteorology (distance measurement) [53], spectroscopy [54], microwave photonics [55], quantum computing, and human breath analyzer for monitoring diseases [56].

The original method for generating OFC involves using a mode-locked laser with a stable frequency and repetition rate. Mode-locked lasers generate periodic pulse trains in the time

domain that converts into the frequency domain as a spectrum of equally spaced frequencies [57]. The frequencies of each comb line ($\nu_n$) can be expressed as:

$$\nu_n = f_{\mathrm{CEO}} + n f_{\mathrm{rep}} \tag{2.0}$$

where $f_{\mathrm{CEO}}$ is the carrier-envelope offset frequency, $f_{\mathrm{rep}}$ is the repetition frequency of the optical pulse, and $n$ is the order of OFC. Fig. 2.1 illustrates an example of an OFC generated using a mode-locked laser [58]. The spectrum of the OFC and its optical pulse sequence satisfy the Fourier transform. In the time domain, the relative phase between the carrier wave (represented by a thick red line) and the pulse envelope (solid blue line) changes from pulse to pulse by $\Delta\phi_{\mathrm{CEO}}$ leading to a non-zero carrier-envelope offset (CEO) frequency in the frequency domain, causing the comb spectrum to shift away from the origin. The carrier-envelop offset frequency ($f_{\mathrm{CEO}}$) is the offset between the first comb tooth and the frequency origin (zero frequency). It arises from the phase difference $\Delta\phi_{\mathrm{CEO}}$ and establishes a link between the frequency and time domains through the Fourier transform. The term $f_{\mathrm{CEO}}$ represents an overall frequency shift of the entire optical frequency comb spectrum from the frequency origin. The comb spacing of the OFC corresponds to the laser's repetition rate ($f_{\mathrm{rep}}$) in the frequency domain [58].




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**Figure 2.1:** Mode-locked laser-based optical frequency comb representation. (a) In the time domain, a pulse train from a mode-locked laser shows slippage $\Delta\phi_{\mathrm{CEO}}$ between the carrier phase (thick red line) and the pulse envelope (solid blue line) from one pulse to the next. (b) In the frequency domain, this phase slippage $\Delta\phi_{\mathrm{CEO}}$ leads to the carrier-envelope offset frequency from the Fourier transform between the time and frequency domain. The frequency repetition rate ($f_{\mathrm{rep}}$) represents the spacing between comb lines [taken from Schilt et al. [58]]. © 2015 by the authors; licensed under the terms and conditions of the Creative Commons Attribution license (http://creativecommons.org/licenses/by/4.0/)

Various techniques have been developed for generating OFC, for example, based on cascaded modulators [59] and nonlinear fiber mixing [60]. Recent advancements focused on an integrated approach to develop a compact and efficient OFC laser with low power consumption [61]. Various materials are employed to achieve efficient nonlinear effects. However, SiN material appealed to the researcher due to its moderate third-order nonlinear coefficient [25], low intrinsic loss [11], negligible two-photon absorption (TPA) coefficient [62], and high optical power handling. One of the effective methods for

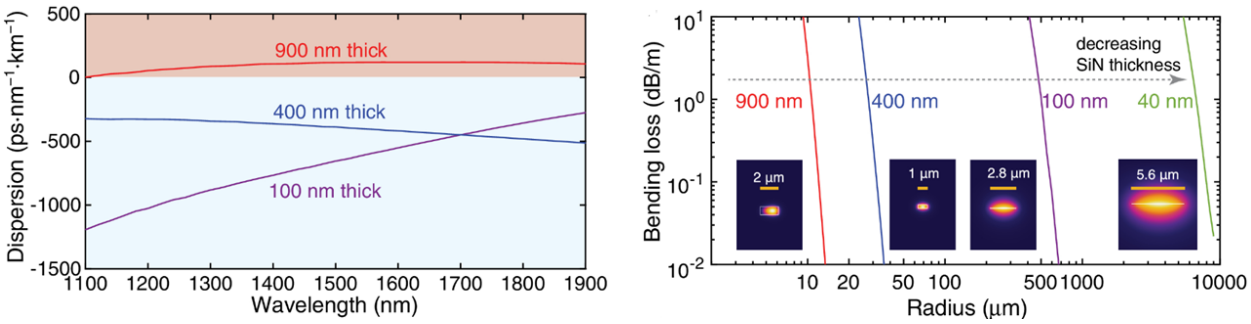
developing an on-chip OFC laser is a SiN microring resonator. SiN microring provides a high-quality factor with low loss. A quality factor (Q) of a resonator is the measure of the sharpness of the resonant peak. It is defined as the ratio of the resonating wavelength over the its spectral width. A high Q ring resonator provides high optical confinement (narrow linewidth) within the cavity for longer, to enhance nonlinear efficiency [63].

## 2.2 Silicon Nitride Microring for OFC Generation

Significant progress has been achieved in advancing integrated nonlinear photonic devices across various platforms over the last few decades [64]. In the early stages, OFC generations were fabricated in silica [20] and silicon [21]. However, these platforms suffer from nonlinear loss mechanisms, notably a high TPA coefficient in silicon. This challenge prompted the transition of comb generation technology to the SiN platform, which offers several advantages. SiN technology enables wide comb spacing from the mid-ultraviolet (400 nm) to mid-infrared (3.5 µm), negligible two-photon absorption [65], high optical power handling capability, low linear loss, the ability to fabricate ultra-high Q cavities [66], and high third-order nonlinearity ($\chi^{(3)}$). These properties make SiN material ideal for microresonator-based OFC generation [67].

Recent studies demonstrate that a broadband OFC laser can be achieved in a zero or anomalous waveguide dispersion. The dispersion characteristics of SiN waveguides are influenced by their dimensions, including both width and thickness. To achieve anomalous

dispersion, a thicker SiN film is required. Whereas thin SiN film devices typically operate within the normal dispersion regime. The maximum nonlinear effect can be achieved in a sufficiently thick silicon nitride film in a dispersive engineering device to achieve anomalous dispersion [68–70] with experimentally demonstrated comb generation [71, 72]. A third-order parametric nonlinear optical process known as four-wave mixing (FWM) can also generate a broadband OFC laser from a single laser source in an anomalous dispersion micro-ring resonator [73, 74], leading to an efficient solution.




(a) (b)

**Figure 2.2:** (a) Waveguide dispersion for varying SiN core thickness, blue refers to normal dispersion ($D < 0$) and red refers to ($D > 0$); (b) SiN waveguide bending loss versus bending radius of the TE0 optical mode profile for different SiN core thicknesses (taken from [28] with copyright permission from the Photonics Research Journal).

Fig. 2.2(a) plots the dispersion of different SiN thicknesses over wavelengths. It shows that SiN core thickness of 900 nm operates within the anomalous dispersion region, while thinner SiN cores (100 nm and 400 nm) exhibit significant normal dispersion [28]. A large normal dispersion suppresses the nonlinear effects due to its spectral broadening feature.

The normal dispersion of thin SiN film can be converted to anomalous dispersion using a dispersive engineered waveguide or photonic crystal structures [75]. However, designing such waveguides to exhibit normal dispersion to anomalous dispersion is challenging due to limited variable parameters, poor light confinement, increased fabrication challenges, and additional losses in the waveguide. Fig. 2.2(b) shows the bending loss of 900 nm, 400 nm, and 100 nm thick SiN core waveguides. In terms of mode confinement and footprint, 900 nm thick SiN film provides good confinement leading to a small footprint due to a lower bend loss at a small radius. Therefore, researchers use a thicker SiN layer as a core waveguide for achieving anomalous dispersion that needed for OFC generation [76]. However, the fabrication of thicker SiN films is prone to cracking due to fabrication-induced stress on the lower layers [77]. This crack decreases the Q factor of the microring resonator due to the additional losses [78]. Therefore, in this thesis, the microring design methodology first focuses on dispersion engineering of the ring waveguide geometry (width and radius) to reduce the normal dispersion to the desired level. Once the target dispersion characteristics are achieved, the ring radius is selected to minimize bending loss at the chosen width. This sequential design approach effectively balances dispersion control with optical confinement, ensuring low bending loss.

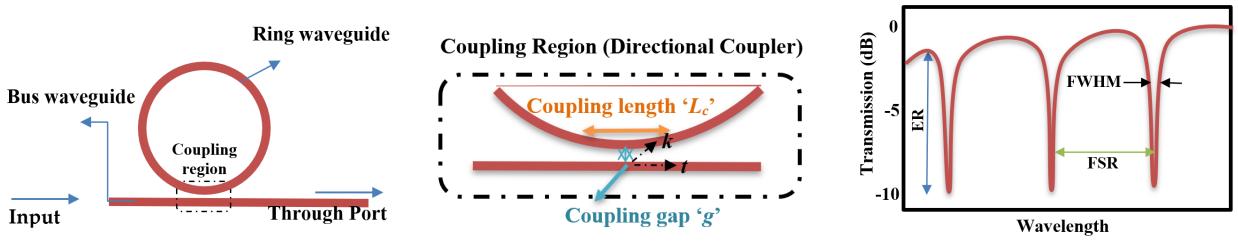
The fabrication challenges mentioned above motivate the investigation of OFC in a SiN ring resonator with large normal dispersion. Most of the publications show numerical analysis for generating OFC in normal dispersion. Recent research developments have

demonstrated OFC generation using a 550 nm thick SiN microring resonator. This design utilizes microheater actuation to control the comb spectrum, with a ring radius of 100 µm and a comb line spacing of 231 GHz [79]. The microheater enables precise and stable control of the SiN microring's resonance wavelength. This allows for finer detuning of the pump resonance instead of detuning the pump laser wavelength directly as conventionally done [79]. Another study demonstrated a dual coupled microring using mode coupling method to generate comb lines with different FSR [80, 81]. This mode coupling method generates a comb spectrum in normal dispersion using the nonlinear coupled-mode equation. Mode coupling occurs between two cavity modes with different dispersions, resulting in FSR changes for each mode. When the respective FSR of two modes are aligned at a resonating wavelength, mode coupling is reached at which point, normal dispersion shifts to anomalous dispersion, generating an OFC at this coupling wavelength [81]. In this reported work, the cross-sectional dimensions of the SiN microring are 2 µm in width and 550 nm in thickness and the design requires a large footprint and high input power, up to 1.75 W. Another study demonstrated a high-power comb spectrum generated using a coupled ring geometry with a programmable heater for precise wavelength control and mode-crossing avoidance [82]. The SiN core waveguide in each ring had a cross-sectional dimension of 1000 nm in width and 730 nm in thickness. It achieved up to 41% pump-to-comb conversion efficiency with 180 mW input power in normal dispersion. However, the design suffers from high ring coupling loss, high power

requirement, and integration complexity. The FWM effect has also been explored in normal dispersion SiN microring resonators by adding tellurium-oxide ($TeO_2$) material as the top cladding [83]. The researcher demonstrated that increasing $TeO_2$ cladding thickness reduces the magnitude of normal waveguide dispersion and increases FWM efficiency [83]. Adding other materials to the fabrication process, however, creates complexity and raises device manufacturing costs. This thesis focuses on the potential of OFC generation in SiN microring resonators using FWM within normal dispersion regions from a thinner SiN layer that is more fabrication friendly.

### 2.2.1 Microring Resonator Principles

A single-bus microring resonator is an integrated micro-scale device that propagates light into a circular waveguide and receives the light from a coupled straight (bus) waveguide at a specific wavelength.Fig. 2.3(a) shows a single-bus microring resonator. The microring resonates at specific wavelengths and the maximum amplitude of the electric field propagates inside the cavity at those wavelengths.




(a) (b) (c)

**Figure 2.3:** (a) Single-bus microring resonator; (b) basic design of the ring directional coupler ($t$: transmission coupling, $k$: field coupling)(c) normalized optical transmission response (ER: extinction ratio, FSR: free-spectral range, FWHM: full wave half maximum)

The constructive interference inside the cavity creates resonances when the optical path length of the resonator is equal to a round-trip phase shift in the ring equal to $2m\pi$ where $m$ is an integer [84].

$$\beta \cdot L = 2m\pi \quad \text{where} \quad m = 1, 2, 3, \ldots \tag{2.1}$$

where $L$ is the ring circumference of the optical path. The optical path length and the resonant wavelength are key parameters to design a ring resonator with low bend loss and a small free spectral range (FSR). A small FSR provides more resonances within a specific optical bandwidth and also enhances the interaction between modes, increasing the nonlinear effects in the microring.

A racetrack resonator consists of two straight waveguide sections connected by two semi-circular bends, forming a shape that resembles a racetrack instead of a simple circular ring.

If the ring is a racetrack, the path length $L$ is given by $L = 2\pi R + 2L_c$, where $R$ is the radius of the semi-circular bend and $L_c$ is the coupling length of the straight waveguide in the coupling region.

The propagation constant $\beta$ is defined by

$$\beta = \frac{2\pi n_{eff}}{\lambda_r} \tag{2.2}$$

where, $n_{eff}$ is the effective refractive index, $\lambda_r$ the resonating wavelength. Equation (2.2) is used in (Eq. 2.1) to determine the resonating wavelengths of the microring.

$$\lambda_r = \frac{n_{eff}L}{m} \tag{2.3}$$

An important part of the ring resonator is its directional coupler (DC) that couples light between the bus waveguide and the ring waveguide. A DC is designed with a specific coupling gap and a coupling length (Fig. 2.3(b)). A DC coupler design can operate under three possible coupling regimes: under-coupling, over-coupling, and critical coupling. In the under-coupled regime, the ring waveguide is positioned at a relatively large distance from the bus waveguide in the coupling region. In this case, the light is not fully coupled into the ring waveguide through the coupling region, which makes the coupling loss higher than the intrinsic loss of the ring. On the other hand, in the over-coupled regime, the ring waveguide is placed closer to the bus waveguide. Under this condition, more light is coupled into the ring waveguide,

causing the coupling loss lower than the intrinsic loss of the ring. The critical coupling is achieved when the coupling loss is identical to the intrinsic loss of the ring. It provides a high extinction ratio with minimum transmission light $T_{\text{min}}$ at resonance. To achieve a high extinction ratio at the resonating wavelength, the design in this thesis employs the critical coupling condition of the microring resonator. In a lossless ring resonator, light is conserved such that the following equation holds.

$$k^2 + t^2 = 1 \tag{2.4}$$

where $k$ represents the field coupling coefficient and $t$ represents the transmission coupling coefficient. The field coupling coefficient $k$ is given by

$$k^2 = \frac{P_{\text{coupled}}}{P_{\text{in}}} = \sin^2(C \cdot L_c) \tag{2.5}$$

where $P_{\text{in}}$ is the input optical power to the ring bus waveguide, $P_{\text{coupled}}$ is the optical power coupled to the ring waveguide through the directional coupler, $L_c$ is the coupling length of the directional coupler, and $C$ is the coupling coefficient of the directional coupler. The coupling coefficient $C$ is expressed as

$$C = B \cdot e^{-A \cdot g} \tag{2.6}$$

where $A$ and $B$ are constant values dependent on the geometry of the directional coupler (*i.e.*, waveguides dimensions and refractive indices) and the optical wavelength. The coupling gap $g$ is the lateral distance between the bus waveguide and ring waveguides [85]. Using (Eq. 2.6), the coupling coefficient $C$ can be determined through numerical simulation for different coupling gaps. Using (Eq. 2.4), the transmission coupling coefficient $t$ in a lossless ring resonator is simply defined as,

$$t^2 = \frac{P_{\text{through}}}{P_{\text{in}}} = \cos^2(C \cdot L_c) \tag{2.7}$$

where $P_{\text{through}}$ is the optical power transmitted through the bus waveguide after the directional coupler.

The transmission response of the ring resonator (Fig. 2.3(c)) is obtained by solving the field amplitude $E_{\text{thru}}$ corresponding to the optical signal in the waveguide bus after the DC at the through-port and is a function of wavelengths and phase.

$$E_{thru} = \frac{te^{-j\phi} - \alpha}{e^{-j\phi} - \alpha t^*} \tag{2.8}$$

where $\alpha$ is the linear waveguide propagation loss coefficient, $t$ is the transmission coupling coefficient (Eq. 2.7) with $t^*$ as its complex conjugate. The mode-dependent factor $\phi$ is the linear phase shift of the propagating light in the ring resonator (Eq. 2.1).

The characteristics of the transmission response in Fig. 2.3(c) depend on losses, the

effective index of the propagating optical mode, the ring resonating wavelengths, and the directional coupling coefficient [87]. The extinction ratio (ER) provides the difference in optical power between the optical transmission off-resonance ($T_{max}$) and at resonance ($T_{min}$):

$$ER = \frac{T_{max}}{T_{min}} \tag{2.9}$$

The free spectral range (FSR) is the distance between two resonating wavelengths ($\lambda_{r_{n+1}} - \lambda_{r_n}$, $n$ being an integer) and can be calculated as:

$$FSR = \frac{\lambda_r^2}{n_g L} \tag{2.10}$$

where $\lambda_r$ is one of the resonating wavelengths that exhibits maximum extinction ratio and $n_g$ is the group index of the waveguide [88]. The group index is defined as $n_g = n_{eff} - \lambda \frac{\partial n_{eff}}{\partial \lambda}$. The FSR as defined in (Eq. 2.10) indicates that it is inversely proportional to the effective refractive index and the ring's circumference, while being proportional to the resonating wavelength square [89].

The laser comb spacing line of the generated OFC is the distance between two adjacent comb lines. This wavelength spacing can either match the FSR of the microring resonator or deviate from it, depending on the adjustments made to the pump power and the detuning, the latter being the wavelength alignment between the pump source and ring resonating wavelengths. The spacing also depends on dispersion [79].

The optimal optical pump power needs to avoid high-order nonlinearities in the waveguide such as FWM and stimulated Brillouin scattering (SBS) that can cause complex interactions between the pump waves. These interactions lead to distortions in the frequency comb and cause the comb line wavelength spacing to deviate from the ring's FSR. Fine detuning between the pump laser wavelength and the ring's resonance wavelengths lead to phase matching required for equally spaced comb lines. Another challenge is that while employing nonlinear effects in the microring, the large dispersion (normal) increases the effective refractive index to change with wavelength, causing a shift in the resonating wavelengths from their original values. As a result, the comb spacing deviates from the FSR of the microring. [90].

A quality factor (Q) of a resonator measures the sharpness of the resonance. It is defined as the ratio of the operating wavelength over the resonance spectral width [91]. The microring resonator Q is defined from two sources. First, the loaded quality factor ($Q_L$) of the ring resonator is a measure of the spectral resonance sharpness at its resonating wavelength:

$$Q_L = \frac{\lambda_r}{FWHM} \tag{2.11}$$

where FWHM is the full width at half maximum which is the spectral width of the resonance 3 dB below the maximum microring transmission (Fig. 2.3(c)). Second, the intrinsic or unloaded quality factor ($Q_i$) only accounts for the ring's propagation and scattering losses

(not consider the direction coupler) defined as:

$$Q_i = \frac{\lambda_r}{\alpha_P \cdot R \cdot FSR} \tag{2.12}$$

where $R$ is the radius of the ring, $\alpha_P$ is the linear propagation loss per unit length of the ring in $\text{cm}^{-1}$. The often reported loss per unit length in dB/cm is converted into its linear unit correspondence ($\alpha_P$ [dB/cm] $= 4.34 \times \alpha_P$ [$\text{cm}^{-1}$]).

### 2.2.2 Nonlinear Optics

Nonlinearity in optics is a phenomenon where a change in the material optical properties occurs with high optical power. Specifically, the refractive index changes leading to changes in the phase of propagation light. Such phenomenon allow to generate second or third harmonics of the fundamental sources through FWM and other nonlinear optical processes, such as self phase modulation (SPM) and cross phase modulation (XPM) [92].

In this thesis, an analytical approach is used to investigate the optical comb generation with FWM through numerical simulation of the nonlinear Schrodinger equation (NLSE) [93]. The NLSE derivation consists of both linear and nonlinear parameters of the waveguide and source wavelength. The split-step Fourier method is then used to solve the NLSE equation leading to the following expression defining the electric field $\boldsymbol{E}$ propagating in an optical

waveguide with the propagation in the z direction [94].

$$\frac{d\boldsymbol{E}}{dz} = -\frac{\alpha}{2}\boldsymbol{E} - \frac{i\beta_2}{2}\frac{d^2\boldsymbol{E}}{dt^2} + i\gamma|\boldsymbol{E}|^2\boldsymbol{E} \tag{2.13}$$

where $\alpha$ is the optical linear loss coefficient of the waveguide, $\beta_2$ is its second order group velocity dispersion (GVD), and $\gamma$ is the waveguide nonlinear coefficient. An important parameter in (2.13) is the required dispersion to achieve efficient nonlinear properties in a dielectric medium. Different frequencies of light propagate at different speeds in an optical waveguide, resulting in pulse broadening known as waveguide dispersion caused by the frequency dependence of the refractive index [95]. This dispersion causes the propagating optical pulse to broaden and reduces the nonlinear effects if the dispersion is left uncontrolled. Dispersion, which can be of three types (anomalous, zero, and normal, which will be defined shortly) is the determinant characteristic of a waveguide that controls nonlinear effects. The waveguide dispersion ($D$) is expressed as [96]:

$$D = \frac{d\beta_1}{d\lambda} = -\frac{2\pi c}{\lambda^2}\beta_2 \tag{2.14}$$

where $c$ is the speed of light, $\lambda$ is the operating wavelength, and $\beta_1$ is the first-order dispersion corresponding to the first derivative of the propagation constant as a function of wavelength.

$\beta_1$ corresponds to the inverse of the group velocity and can be expressed as:

$$\beta_1 = \frac{1}{v_g} = \frac{c}{n_g} = \frac{c}{n_{\text{eff}} - \lambda \frac{\partial n_{\text{eff}}}{\partial \lambda}} \tag{2.15}$$

The term $\beta_2$ is the second derivative of the propagation contact $\beta$ as a function of the angular frequency and is referred to as the group velocity dispersion (GVD):

$$\beta_2 = -\frac{d^2\beta}{d\omega^2} = -\frac{\lambda^3}{2\pi c^2}\frac{d^2 n_{\text{eff}}}{d\lambda^2} \tag{2.16}$$

where $\beta = \frac{2\pi n_{\text{eff}}}{\lambda}$ is the propagation constant.

The pulse broadening during propagation is closely tied to the GVD amount and type. Pulse broadening occurs due to the different propagation speeds of the individual wavelength components that make up the pulse. This dispersion can be categorized as normal, where GVD is positive $\beta_2 > 0$ and longer wavelength travels faster. On the other hand, shorter wavelength travels faster in anomalous dispersion, where GVD is negative $\beta_2 < 0$ [96]. Zero dispersion occurs at a specific wavelength where GVD is zero ($\beta_2 = 0$). One of the effective methods to achieve zero dispersion in an optical system is by using dispersion shifting fiber [97]. Normal dispersion causes temporal broadening of the pulse as it propagates. Anomalous dispersion causes compression rather than broadening of a pulse.

The nonlinear coefficient $\gamma$ determines the strength of nonlinear optical effects in a dielectric medium and quantifies the optical intensity impact on its propagation properties

from nonlinear interactions. The nonlinear coefficient $\gamma$ is given by:

$$\gamma = \frac{2\pi n_2}{A_{\text{eff}}\lambda} \tag{2.17}$$

where $n_2$ is a measure of the nonlinear refractive index of the medium [98], $A_{eff}$ is the effective mode area of the optical mode in the core of the waveguide, and $\lambda$ is the operating wavelength.

### 2.2.3 Four-Wave Mixing

The third-order parametric nonlinear optical process known as FWM can be used to generate an efficient OFC laser at relatively low input power. FWM is a nonlinear process where two optical pump signals interact inside a dielectric medium when they have the same wavelength (degenerate FWM) or different wavelengths (non-degenerate FWM). The FWM phenomenon is also known as the Kerr nonlinear effect when the high optical power is injected inside the microring generating two wavelength sidebands in the form of an idler wavelength and a signal wavelength [99]. The generated wavelength sidebands are equidistant following both the energy and the momentum conservation laws. This FWM process continues in the ring and forms the OFC spectrum.

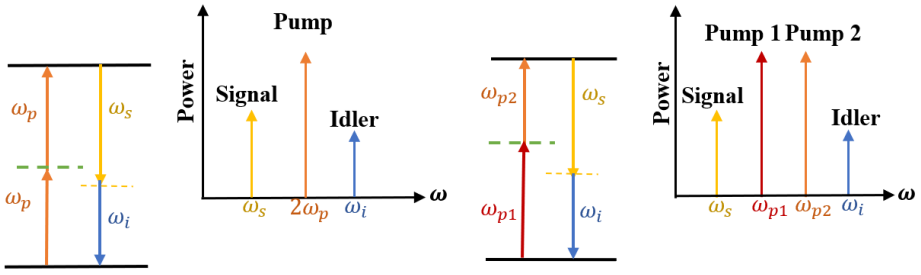



**Figure 2.4:** Conceptual illustration of the FWM nonliear process using the energy level band diagram and the corresponding spectral of the wavelength conversion leading to an idler ($\omega_i$) and signal ($\omega_s$): (a) de-generate FWM when the optical pump signals have the same frequency ($\omega_p$); (b) non-degenerate FWM when their frequencies are different ($\omega_{p1}$ and $\omega_{p1}$).

A high-power continuous-wave (CW) pump laser can be used to initiate the two pump photons ($\omega_p$) at the same frequency and generate a signal photon ($\omega_s$) and an idler photon ($\omega_i$) at new wavelengths in the dielectric medium (Eq. 2.18). This phenomenon is known as degenerate four-wave mixing (FWM) and is shown in Fig. 2.4(a).

$$2\omega_p = \omega_s + \omega_i \tag{2.18}$$

When two CW lasers initiate two pump photons at different frequencies ($\omega_{p1}$ and $\omega_{p2}$) in the dielectric medium, it excites a signal photon and idler photon at new wavelengths. This phenomenon is known as non-degenerate FWM, as shown in Fig. 2.4(b). The relation between all wavelengths is given by:

$$\omega_{p1} + \omega_{p2} = \omega_s + \omega_i \tag{2.19}$$

The efficiency of FWM is influenced by the optical intensity, $I$, of the interacting pump waves. The dielectric's refractive index $n$ changes with light intensity. The correlation between the changing refractive index of the medium from the light intensity is given by:

$$n = n_0 + n_2 I \tag{2.20}$$

where $I$ is the intensity of the pump source, defined as the ratio of the input pump power to the effective mode area of the waveguide. As the light intensity increases, the refractive index increases due to the nonlinear term $n_2 I$. This intensity-dependent change in the refractive index affects the phase of lightwaves and influences the efficiency of nonlinear optical processes like four-wave mixing (FWM) [100].

### 2.2.4 OFC Generation in Microring using Non-degenerate FWM

Optical frequency comb (OFC) generation in a microring resonator can be achieved via the non-degenerate four-wave mixing (FWM) process. This process requires a balance between the nonlinearity and dispersion of the ring waveguide, as well as a balance between optical gain and loss, as described by NLSE (Eq. 2.13), which governs light propagation in the microring cavity [101]. A high-Q microring resonator is also essential to sustain light

circulation over many round trips, enabling constructive interference and maintaining the injected pump power within the cavity for a prolonged time.

In the non-degenerate FWM technique, two pump waves at distinct wavelengths, either with equal or unequal powers, are coupled into the microring. Their nonlinear interaction generates signal and idler sidebands at $\omega_s$ and $\omega_i$, satisfying the energy conservation relation (Eq. 2.19). FWM results in the redistribution of energy from the pump mode to the two primary sidemodes. When the intracavity pump power surpasses the FWM threshold, energy is efficiently transferred from the pumps to these first-order sidebands over the multiple round trips, and the cascaded FWM process appears. The FWM threshold is defined as the minimum intracavity power at which the nonlinear optical interactions enable the generation of sidebands through four-wave mixing. In this process, two of the pump modes and two side modes with frequencies $\omega_s$ and $\omega_i$ transfer to another two modes with frequencies $\omega_p$ and $\omega_q$, satisfying $\omega_s+\omega_i= \omega_p+ \omega_q$ [102]. This process cascades until an equilibrium state is reached, producing a coherent, evenly spaced optical frequency comb.

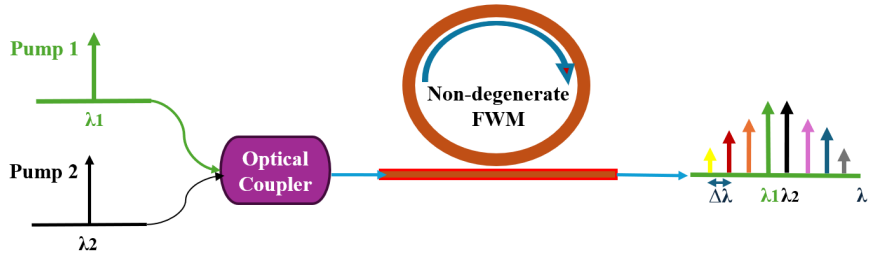



**Figure 2.5:** Illustration of OFC generation in a microring resonator through non-degenerate FWM..

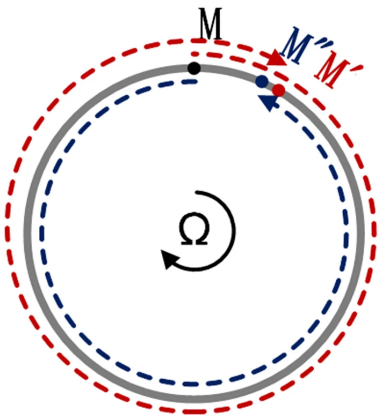
The comb spacing ($\Delta\lambda$) refers to the wavelength separation between adjacent comb lines, as illustrated in Fig. 2.5. Ideally, the comb spacing in a microring resonator is equal to its FSR. However, in the presence of high normal dispersion in the microring, the actual comb spacing can deviate from the nominal FSR [79]. The number of comb lines and their corresponding optical power depend on the optical pump power, cavity $Q$-factor, waveguide dispersion, and nonlinear coefficient. Higher pump powers combined with optimized dispersion conditions can extend the comb bandwidth and enhance power transfer into the sidebands.

This thesis investigates non-degenerate four-wave mixing (FWM) in a SiN racetrack microring for OFC generation. Through this process, equidistant sidebands are produced, with all frequency components maintaining equal spacing. This equidisant occurs because the signal and idler sidebands inside the cavity exhibit similar power levels to the pump photons, resulting in a uniformly spaced OFC.

## 2.3 Optical Gyroscope

Gyroscopes are crucial in positioning, navigation, and timekeeping (PNT) solutions for enhanced global navigation satellite system (GNSS) navigation accuracy. An optical gyroscope is a sensor designed to measure rotational movement using light principles. The operational principle of an optical gyroscope depends on the Sagnac effect [103] in which a CW laser beam is divided by a beam splitter into two beams that propagate in opposite

directions around a circular integrated waveguide or an optical fiber. With rotation, the optical path lengths for the two counter-propagating beams become unequal resulting in a phase shift between the two beams. Fig. 2.6 illustrates the principle with the two light beams entering at point M and propagating through a ring waveguide (represented by the solid gray line). The illustration compares the behavior of the light beams when the ring waveguide is stationary and when it is rotating. When stationary, the two beams traverse identical optical paths and return to point M, simultaneously with no phase difference between the beams. When the waveguide undergoes clockwise rotation relative to the plane of the optical beams, the clockwise beam (denoted by the red dashed line) experiences an extended effective optical path length compared to the counterclockwise (CCW) beam (denoted by the blue dashed line). Consequently, the clockwise beam arrives at point M' at a different time than the CCW beam (at point M"). This path difference results in a phase shift between the two beams, which is used to quantify the rotation rate of the device [104].




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**Figure 2.6:** Illustration of the Sagnac effect in a ring waveguide. When stationary, two light beams injected at point M follow identical optical paths. In a clockwise rotation, the clockwise beam (red dashed line) experiences a longer effective optical path (displacing point M to M') compared to the counterclockwise (CCW) beam (blue dashed line) displaced to M" [104]. (This open-access article is distributed under the terms of the Creative Commons Attribution license (http://creativecommons.org/licenses/by/4.0/)

Over the last decades, high-precision gyroscopes, such as ring laser gyroscopes (RLGs) [105] and interferometric fiber optic gyroscopes [106] (IFOGs) based on Sagnac effects, have been predominantly used. However, RLGs are prone to mechanical vibrations [107] due to lock-in effects, while IFOGs experience a significant bending loss due to the extended fiber length in their cavities [108]. The lock-in effect in an RLG is a phenomenon where the gyroscope cannot detect rotational motion below a low rate threshold. In this lock-in condition, the frequency difference between the clockwise and counterclockwise (CCW) light beams becomes negligible, leading to insensitivity to

rotational changes [109]. These problems lead to challenging development related to increased weight, larger size, and higher power consumption. To overcome these limitations, chip-scale integrated optical gyroscopes (IOGs) have gained research attention due to their compact design [110], lower cost [111], and resiliency to vibrations [112]. Various materials have been explored for developing IOGs. Silicon nitride (SiN)–based IOGs showed improved accuracy from low waveguide propagation loss and noise suppression from polarization dependent loss [17, 49]. Despite these advancements, IOGs can still exhibit low sensitivity under unbiased conditions and the absence of rotation [113]. Biasing the IOG using an external phase shifter or modulation allows to enhance the gyroscope's sensitivity by shifting the baseline of the rotation-induced phase difference (Fig. 2.6) at the steeper slope of the raised cosine function of the rotation-induced phase difference [114].

## 2.4 State-of-arts of PZT on SiN Phase Modulator

PIC development using SiN material leverages its linear properties as well as its ultra-low propagation loss, low-temperature sensitivity, and low refractive index contrast [115]. These features improve the performance in diverse applications, including optical gyroscopes, LiDAR, tunable lasers, and data center interconnects [116, 117]. SiN can be developed as active devices by integrating electro-optic thin film materials (graphene, lithium niobate) [118] enabling tunable components such as phase shifters, modulators,

actuators, and switches. Despite notable advancements in integrated actuators that convert energy into mechanical force, inherent limitations hinder the usefulness of reconfigurable PICs due to their relatively complex hybrid integration and high optical losses of the electro-optic thin films used in the actuator material. The development of hybrid integration of electro-optic thin films with SiN material for signal modulation is inherently complex [119]. Further, direct electro-optic modulation in the actuator has high loss, up to 5 dB, with currently used material (*i.e.*, graphene [120], PZT [121], or $LiNbO_3$ [118]).

The most prevalent integrated modulation method for gyroscopes uses thermo-optic heating elements, in which joule heating in a metal strip induces refractive index change in the waveguides [122]. The thermo-optical effect, however, is characterized by low tuning speed (1 ms) and high electrical power consumption ($>$ 300 mW) [123], limiting their use for large-scale integration. To address these challenges, recent research investigates stress optic effects in modulating SiN devices or micro-electro-mechanical systems (MEMS) technology [124–126]. For stress-based modulation, the piezoelectric material $PbZr_xTi_{1-x}O_3$ (PZT) stands out with a high piezoelectric coefficient which quantify the amplitude of the electric field generated by a force applied to a material [127]. This property allows the piezoelectric material such as PZT to deform the waveguide structure from a mechanical stress generated by applying a voltage to a metal, as shown in Fig. 2.7. An actuator using a PZT material (green section in Fig. 2.7) between two metal electrode layers (yellow sections) on top of a SiN waveguide in red surrounded by a silica cladding in

light blue. When a voltage is applied across the electrodes, an electric field is generated in the PZT layer. The electric field causes deformation of the PZT material through the inverse piezoelectric effect. This deformation results in a mechanical stress (indicated by red arrows) within the PZT actuator, which then propagates through the SiN waveguide.

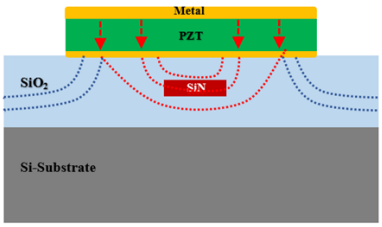



**Figure 2.7:** Cross-section schematic illustrating the stress (red straight lines between the two metal electrodes) in an optical modulator using a PZT actuator above a SiN waveguide surrounded by silicon dioxide $SiO_2$ in a silicon (Si) photonic integrated technology platform. The stress comes from the applied voltage between the two metal electrodes causing compressive (blue dashed lines) and tensile stresses (red dashed lines) in SiN waveguide geometry leading to a change in the effective index.

The deformation of the PZT material produces both compressive (negative) and tensile (positive) stresses. These stresses affect the SiN waveguide structure: tensile stress (represented by red dashed lines) expands the SiN core waveguide, while compressive stress (shown by blue dashed lines) affects the $SiO_2$ cladding, further compressing the waveguide structure. These mechanical stresses change the effective refractive index of the waveguide as well as its shape [128] and will be discussed further in Chapter 4. PZT material exhibits potential for high tuning speed operation up to 1 MHz with power consumption as low as 1 µW [128].

PZT-based stress-optic modulators exhibit low loss, and low power consumption (sub-µW) from their small leakage current [129]. Indeed, recent studies have demonstrated the efficiency of PZT-based actuators in SiN photonic devices. For instance, a PZT actuator-based SiN ring modulator with a 1250 µm diameter and 0.03 dB/cm loss leads to 20 nW power consumption, and a 20 MHz electrical bandwidth at an applied voltage of 20 V [130]. Another study showcased a stress-based Mach-Zehnder interferometer (MZI) phase modulator on a passive SiN-based platform, using a PZT layer achieving a power consumption of 300 nW for a $\pi$-phase shift with the modulation speed improved from a short rise time of 253 ns [131]. Further advancements include the integration of a dome-shaped PZT stress actuator with an asymmetric double stripe SiN waveguide enabling modulation and reconfigurable switching, achieving a $\pi$-phase shift at 16 V/cm and a 4.25 µs actuation speed [132]. Aligned with the recent development mentioned above, the work presented in this thesis investigates the improvements enabled by the proposed stress-based phase modulator which integrates a PZT actuator layer on top of a single-mode SiN waveguide for greater sensitivity in an optical gyroscope application.

### 2.4.1 Stress-Optic Effect

Compared to the thermo-optic effect, the stress-optic effect using piezoelectric material provides lower power consumption due to small leakage of current. Indeed, the current can leak through different pathways such as cracks in the PZT material during deposition and

interfacial defects between the PZT material and the metal electrodes [133]. The stress-optic effect is a phenomenon that changes the refractive index of an optical material due to mechanical stress as discuss earlier. Mechanical stress originates from an external or internal electric field generated by the applied voltage and causes deformation in piezoelectric materials. This is known as the piezoelectric effect, which converts electric fields to mechanical deformation and vice versa in a crystalline material [134]. There are two types of piezoelectric effect in the material: inverse effect and direct effect. The inverse piezoelectric effect induces stress in the piezoelectric material when voltage is applied to the electrodes made of metal layers as shown in Fig. 2.8. The voltage generates a vertical electric field, resulting in displacement and mechanical stress in the PZT material [134].

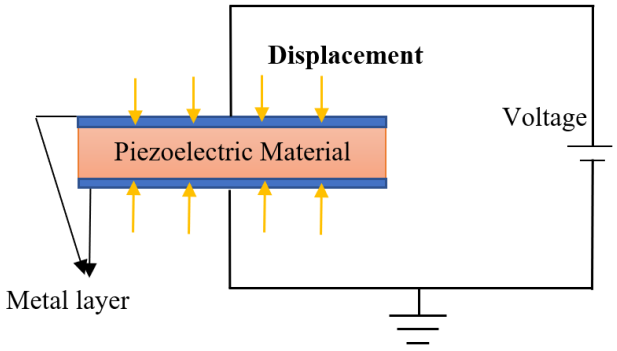



**Figure 2.8:** The PZT actuator operates based on the inverse piezoelectric effect. This causes displacement (illustrated by yellow arrows) in the PZT material, which is sandwiched between two metal layers. The displacement occurs in the PZT layer due to the presence of an applied voltage to the metal layers.

The direct piezoelectric effect, on the other hand, involves the conversion of mechanical energy into electrical energy. When mechanical stress is sensed by a piezoelectric material, it induces an electric field and a corresponding voltage across the material. This thesis investigates the inverse piezoelectric effect as the principle of operation used in the integrated PZT actuator using voltage sources. In the context of optimization, this type offers a simpler and more efficient alternative than integrating ways causing mechanical stress.

The electric field and deformation relationship takes either a strain-charge or a stress-charge form. The stress-charge form is defined by a stress tensor matrix $\boldsymbol{\sigma}$ and a electrical displacement vector $\boldsymbol{D}$ in the PZT material as:

$$\boldsymbol{\sigma} = c_E \cdot \boldsymbol{S} + e^t \cdot \boldsymbol{E} \tag{2.21}$$

$$\boldsymbol{D} = e \cdot \boldsymbol{S} + \varepsilon_s \cdot \boldsymbol{E} \tag{2.22}$$

where, $e$ and $e^t$ are the piezoelectric stress tensors, $\boldsymbol{S}$ is the strain tensor, $c_E$ is the stiffness tensor, $\boldsymbol{E}$ is the electric field vector, $\epsilon_s$ is the dielectric permittivity tensor.

As piezoelectric materials exhibit an anisotropic pattern and the electrical displacement is polarized dependent, the theory requires further understanding of the material properties related to stress, strain, and induced electric field [134]. Anisotropy in piezoelectric materials refers to the variation of the generated electric field with respect to the direction of the applied stress or strain. This directional dependence arises from the asymmetric crystal

structure of the piezoelectric material, which is not equal in all directions and is described by a tensor [135]. As a result, the material exhibits different piezoelectric coefficients along the $x$, $y$, and $z$ crystallographic axes [136].

Next, polarization in piezoelectric materials refers to the alignment of the electric dipoles in the material in the presence of an electrical field, which can be induced through poling [137]. Poling in piezoelectric materials is used to align the electric dipoles within the material in a specific direction. This is achieved by applying a strong external electric field to the material at an elevated temperature. Once the dipoles are aligned, the material is cooled down while maintaining the electric field, locking the dipoles in place. The piezoelectric effect depends on this polarization, as the direction and magnitude of the induced electric charge vary with the direction of the applied mechanical stress. A strong piezoelectric response occurs when stress is applied along the polarization direction, while a weaker or negligible response occurs when stress is applied perpendicular to it [137]. The stress, strain tensors, and the induced electric field, are defined in the $x$, $y$, and $z$ directions. The stress and strain components are typically denoted with indices ranging from 1 to 6 using conventional notation in a matrix form. The matrices describe the relationship between the mechanical and electrical states in piezoelectric materials, illustrating how mechanical stress or strain can induce an electric field and vice versa.

The piezoelectric constitutive equations can be expressed in Voigt notation. This notation simplifies tensor representation in continuum mechanics and materials science by reducing

the number of indices and making equations more manageable for numerical computations. The piezoelectric constitutive equations can be expressed as follow.

$$\begin{bmatrix} \sigma_{xx} \\ \sigma_{yy} \\ \sigma_{zz} \\ \sigma_{yz} \\ \sigma_{xz} \\ \sigma_{xy} \end{bmatrix} = \begin{bmatrix} c_{E11} & c_{E12} & c_{E13} & c_{E14} & c_{E15} & c_{E16} \\ c_{E21} & c_{E22} & c_{E23} & c_{E24} & c_{E25} & c_{E26} \\ c_{E31} & c_{E32} & c_{E33} & c_{E34} & c_{E35} & c_{E36} \\ c_{E41} & c_{E42} & c_{E43} & c_{E44} & c_{E45} & c_{E46} \\ c_{E51} & c_{E52} & c_{E53} & c_{E54} & c_{E55} & c_{E56} \\ c_{E61} & c_{E62} & c_{E63} & c_{E64} & c_{E65} & c_{E66} \end{bmatrix} \begin{bmatrix} S_{xx} \\ S_{yy} \\ S_{zz} \\ S_{yz} \\ S_{xz} \\ S_{xy} \end{bmatrix} + \begin{bmatrix} e_{11} & e_{12} & e_{13} \\ e_{21} & e_{22} & e_{23} \\ e_{31} & e_{32} & e_{33} \\ e_{41} & e_{42} & e_{43} \\ e_{51} & e_{52} & e_{53} \\ e_{61} & e_{62} & e_{63} \end{bmatrix} \begin{bmatrix} E_x \\ E_y \\ E_z \end{bmatrix} \tag{2.23}$$

$$\begin{bmatrix} D_x \\ D_y \\ D_z \end{bmatrix} = \begin{bmatrix} e_{11} & e_{12} & e_{13} & e_{14} & e_{15} & e_{16} \\ e_{21} & e_{22} & e_{23} & e_{24} & e_{25} & e_{26} \\ e_{31} & e_{32} & e_{33} & e_{34} & e_{35} & e_{36} \end{bmatrix} \begin{bmatrix} S_{xx} \\ S_{yy} \\ S_{zz} \\ S_{yz} \\ S_{xz} \\ S_{xy} \end{bmatrix} + \begin{bmatrix} \epsilon_{rS11} & \epsilon_{rS12} & \epsilon_{rS13} \\ \epsilon_{rS21} & \epsilon_{rS22} & \epsilon_{rS23} \\ \epsilon_{rS31} & \epsilon_{rS32} & \epsilon_{rS33} \end{bmatrix} \begin{bmatrix} E_x \\ E_y \\ E_z \end{bmatrix} \tag{2.24}$$

where, $\sigma_{ij}$ represents the matrix elements of the stress tensor components using conventional notation, where $i$ and $j$ are considered six combination of $x$, $y$, and $z$ directions. The stress tensor components are denoted as $\sigma_{xx}$ for the principal stress in the $xx$ direction, $\sigma_{yy}$ for the

principal stress in the $yy$ direction, $\sigma_{zz}$ or for the stress in the $zz$ direction, $\sigma_{yz}$ for the shear stress in the $yz$ direction, $\sigma_{xz}$ for the shear stress in the $xz$ direction, and $\sigma_{xy}$ for the shear stress in the $xy$ direction. Principle stress refers to the normal stress acting perpendicular to the directions of polarization in a piezoelectric material, causing deformation in forms of stretching or compressing. Shear stress refers to the component of stress acting parallel to the direction of polarization in a piezoelectric material, resulting in deformation in forms of sliding layers of the material over each other.

The strain tensor components are represented by $S_{ij}$, while $E_i$ denotes the electric field in the $x$, $y$, and $z$ directions. The elasticity matrix, $c_{ij}$, contains the stiffness coefficients and can be determined by Hooke's Law using the Young modulus and the Poisson ratio of the material. Young modulus is a mechanical property of solid materials that measures stretch and deformation in the material when a force is applied. It is defined as a ratio of applied tensile or compressive stress and the resulting strain in the material [138]. Poisson's ratio measures the deformation of a material in directions perpendicular to the direction of applied force [138]. Finally, the coupling matrix, $e_{ij}$, includes the piezoelectric stress constants, and $\epsilon_{rSij}$ represents the permittivity matrix of the material.

The stress-optic effect causes the refractive index to change in the SiN waveguide material as a function of mechanical stress from the PZT structure as shown in Fig. 2.7. The waveguide is long in the z-direction, so the shear stress tensors ($\sigma_{xy}$, $\sigma_{xz}$, $\sigma_{yz}$) become zero [139]. The relation between the stress tensor and the refractive index change in the material can be

described by the following three equations:

$$\Delta n_x = C_1\sigma_x - C_2(\sigma_y + \sigma_z) \tag{2.25}$$

$$\Delta n_y = C_1\sigma_y - C_2(\sigma_x + \sigma_z) \tag{2.26}$$

$$\Delta n_z = C_1\sigma_z - C_2(\sigma_x + \sigma_y) \tag{2.27}$$

where, $\sigma_x$, $\sigma_y$ and $\sigma_z$ are the principal stress tensor components along the $x$, $y$ and $z$ axis directions, respectively, and $\Delta n_x$, $\Delta n_y$ and $\Delta n_z$ are the material refractive index changes along with $x$, $y$ and $z$ directions, respectively. $C_1$ and $C_2$ are stress-optic coefficients for the material. .

The stress optic coefficients define the relationship between stress and refractive index changes in a specific direction. The stress optic coefficient $C_1$ directly relates the stress to the material's refractive index change in the same direction and $C_2$ accounts for the indirect relationship between the stress in orthogonal directions and the refractive index change [139]. The stress optic coefficients $C_1$ and $C_2$ are dependent on strain optic coefficients and can be derived from:

$$C_1 = \frac{n_0^3\,(p_{11} - 2vp_{12})}{2E_m} \tag{2.28}$$

$$C_2 = \frac{n_0^3\,(p_{12} - v\,(p_{11} + p_{12}))}{2E_m} \tag{2.29}$$

Here, $n_0$ is the linear refractive index of the isotropic optical materials, $E_m$ is Young's modulus, and $v$ is Poisson's ratio of the optical materials. The constants $p_{11}$ and $p_{12}$ are strain-optic constants. An isotropic material exhibits the same material properties independent of direction. The strain optic coefficients of $SiO_2$ are 0.121 for $p_{11}$ and 0.27 for $p_{12}$, measured using birefringence measurement [139]. As there is no direct measurement of the strain-optic coefficient of SiN from the literature [139], we used the strain-optic coefficients of $SiO_2$ for SiN material in this thesis.

### 2.4.2 Phase Change from Mechanical Stress

The phase change in an optical material occurs due to a change in the effective refractive index induced by external factors such as mechanical stress. The effective refractive index influences the phase of the light propagates through the material [140]. When the effective refractive index of the propagating optical signal in the material changes over an optical path length, the phase of the propagating light wave changes. The relation between phase change $\Delta\phi$ and effective refractive index change can be expressed as:

$$\Delta\phi = \frac{2\pi L}{\lambda}\Delta n_{\mathit{eff}} \tag{2.30}$$

where $\Delta n_{\text{eff}}$ is the change in the effective refractive index, $L$ is the optical path length of the device and $\lambda$ is the free-space wavelength of the propagating optical signal. The change in

the effective mode index of the optical signal is defined as the difference between the effective refractive index under mechanical stress $\Delta n_{\text{eff}}(V)$, induced by an external voltage source, and the effective refractive index in the absence of mechanical stress $\Delta n_{\text{eff}}(0)$ [141]. This can be mathematically derived as:

$$\Delta n_{\mathit{eff}} = n_{\mathit{eff}}(V) - n_{\mathit{eff}}(0) \tag{2.31}$$

In this thesis, the effective refractive index of the optical waveguide under mechanical stress is determined by simulating the optical and strain field distribution using the Finite Element Method (FEM) of COMSOL Multiphysics.

# Chapter 3

# SiN Microring Optical Frequency Comb in Normal Dispersion

This chapter presents a silicon nitride (SiN) microring resonator design for an optical frequency comb (OFC) in a waveguide exhibiting normal dispersion. The design approach demonstrates the potential for generating OFC in the C-band through non-degenerate FWM. The chapter covers the simulation, fabrication layout, and experimental results of the SiN microring resonator. The measured effective parameters of the microring are used in a numerical simulation to generate the OFC in normal dispersion. This chapter is organized into four subsections: 3.1) simulated design methodology, including mode profile, dispersion, and transmission response; 3.2) the layout design with each design's function; 3.3) the experimental characterization and a comparison of its performance with other

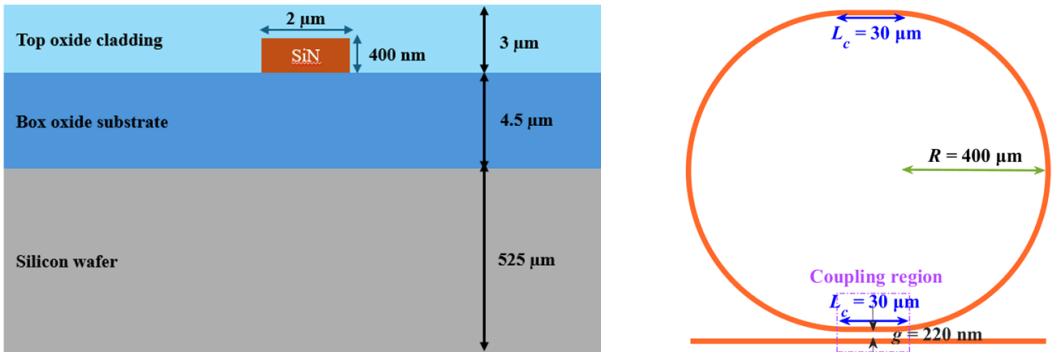
reported designs; 3.4) a discussion on the analytical OFC generation and the simulated spectrum results. I presented some of those results at the Optica Advanced Photonics Congress in July 2024 [142].

## 3.1 Simulated Design Methodology

The simulated methodology for designing a silicon nitride (SiN) microring resonator is conducted in two steps. In the first step, Ansys/Lumerical tools, specifically the Finite Difference Eigenmode (FDE) and 3D Finite Difference Time Domain (FDTD) modules, are employed to design the SiN microring resonator and determine key optical properties such as the effective mode index ($n_{\text{eff}}$), dispersion ($D$), and coupling gap ($g$). In the second step, these optical properties are used in Ansys Interconnect to obtain the transmission response of the SiN microring resonator and assess the design performance. The design focuses on the C band (1530-1565 nm), with a focus on 1547 nm wavelength. This wavelength is recommended in ANT's PDK, where the refractive index is specified and the vertical grating coupler exhibits minimum insertion loss [145]. The cross-sectional view of the SiN waveguide and the design structure of a SiN microring resonator with a 400 µm radius is shown in Fig. 3.1. The thickness of the waveguide's layer is specified by fabrication at Applied Nanotools, Inc. (ANT) [143]. Fig. 3.1(a) illustrates a SiN waveguide that is 2 µm in width and 0.4 µm in thickness, which is placed above a 4.5 µm thick box-oxide layer and surrounded by a 3 µm top oxide cladding. The waveguide structure including the top oxide

cladding and bottom layer is placed on a 525 µm silicon wafer. The width of the SiN waveguide is determined to 2 µm by varying its width from 0.5 µm to 3.5 µm, to reduce the magnitude of normal dispersion from a high value of -2190 ps/nm.km at 1547 nm wavelength. This optimization is crucial as a 400 nm-thick SiN film exhibits significant normal dispersion in the C-band (see Fig. 2.2 in Chapter 2). Fig. 3.1(b) illustrates the ring resonator of racetrack type with an optimized radius of 400 µm. The microring radius is optimized using FDE simulations by sweeping the radius over a range from 30 µm to 400 µm. The resulting optical path achieves a sufficiently small FSR while maintaining low bend loss. Increasing the radius of the resonator decreases the FSR value with an increase in the Q factor leading to higher extinction ratios [144]. The combination of a 400 µm ring radius and a 2 µm waveguide width also significantly reduces normal dispersion. The coupling region of the resonator consists of a 220 nm gap and a 30 µm coupling length which provides a field coupling coefficient of 0.3 through simulation. These features are suitable for generating OFC with low input optical power providing sufficiently high extinction ratios at resonating wavelengths, further discussed later.

The optimized waveguide width of 2 µm ensures an effective fundamental confinement of the TE mode, which is a key requirement to achieve a high Q resonator (Fig. 3.2(a)). The mode profile shows that the optical field is well confined within the SiN core with minimal leakage into the cladding, indicating a strong mode confinement. The design focuses on single-mode light confinement, using refractive indices of 1.9835 for SiN and 1.45 for $SiO_2$,

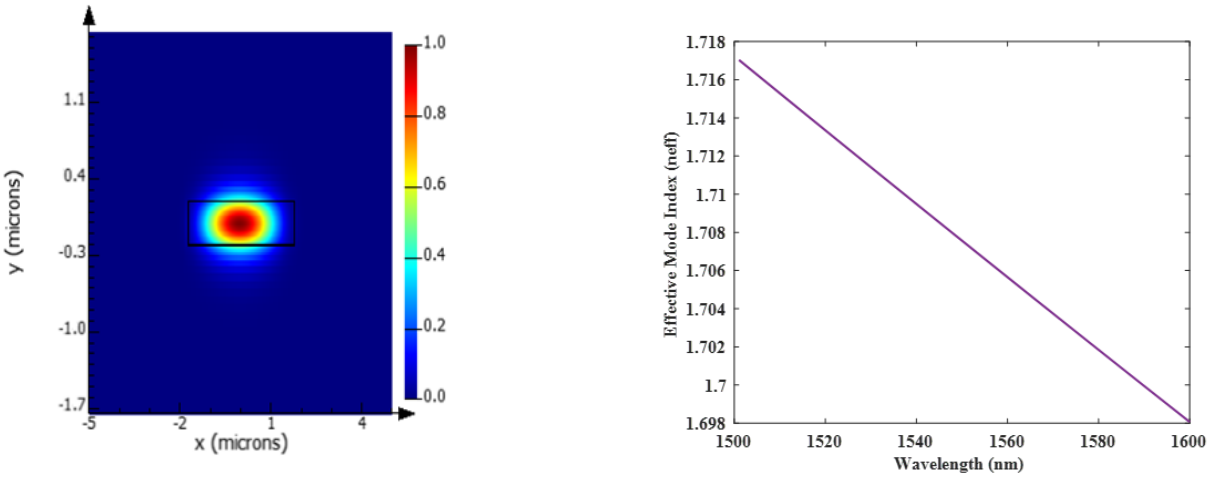



(a) (b)

**Figure 3.1:** (a) Cross-section of SiN technology platform b) Schematic of the 400 µm SiN microring resonator including coupling length ($L_c$) and coupling gap ($g$), and the radius ($R$).

at a wavelength of 1547 nm, as specified by ANT [145]. The effective mode index as a function of wavelength obtained through simulation is shown in Fig. 3.2(b). The effective index decreases as the wavelength increases. The effective refractive index is determined to be 1.7081 at 1547 nm wavelength with a corresponding effective mode area ($A_{eff}$) of 0.722 µm$^2$.

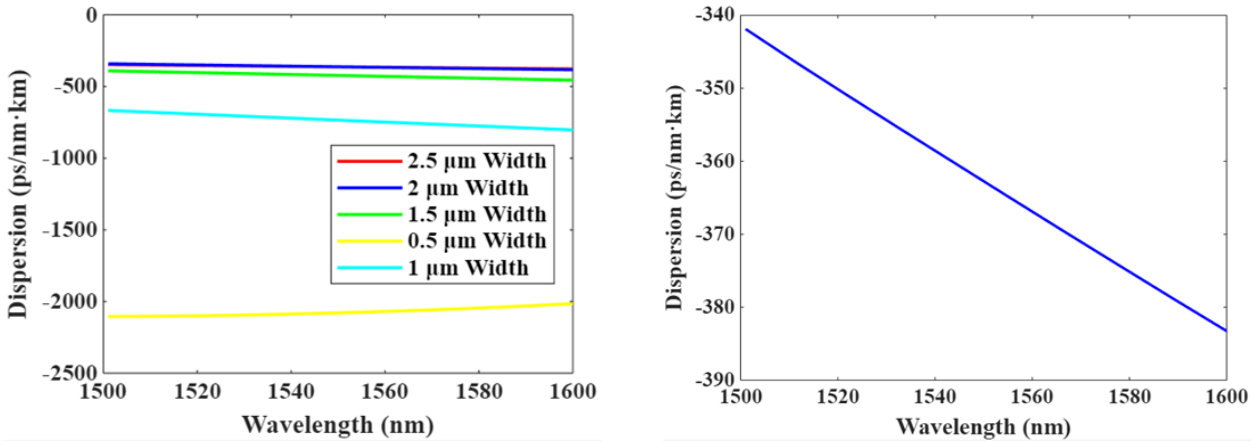
The dispersion curves for the fundamental TE mode (TE0) obtained through simulation are shown in Fig. 3.3. Fig. 3.3(a) compares the normal dispersion (negative dispersion coefficient) for waveguide widths ranging from from 0.5 µm to 2.5 µm, with a fixed SiN microring radius of 400 µm. A waveguide width of 0.5 µm exhibits significant normal dispersion of -2190 ps/nm.km at 1547 nm. As the width increases, the magnitude of normal dispersion decreases, reaching -359 ps/nm.km at 2 µm width. Beyond this width, the dispersion curve flattens, and the 2.5 µm width closely overlaps with the 2 µm case.




(a) (b)

**Figure 3.2:** (a) FDE simulated TE0 mode normalized intensity profile in the SiN waveguide; (b) plot of the effective mode index $n_{eff}$ as a function of wavelength for a 400 nm thick and 2 µm wide cross-sectional SiN waveguide.

Therefore, the 2 µm waveguide width for 400 µm radius microring is selected as optimal design geometry to reduce the normal dispersion. In Fig. 3.3(b), the dispersion of the optimized width of 2 µm increases at longer wavelengths. The resulting dispersion is -359 ps/nm.km at 1547 nm which is targeted in this work to achieve an OFC in more fabrication-friendly normal dispersion.

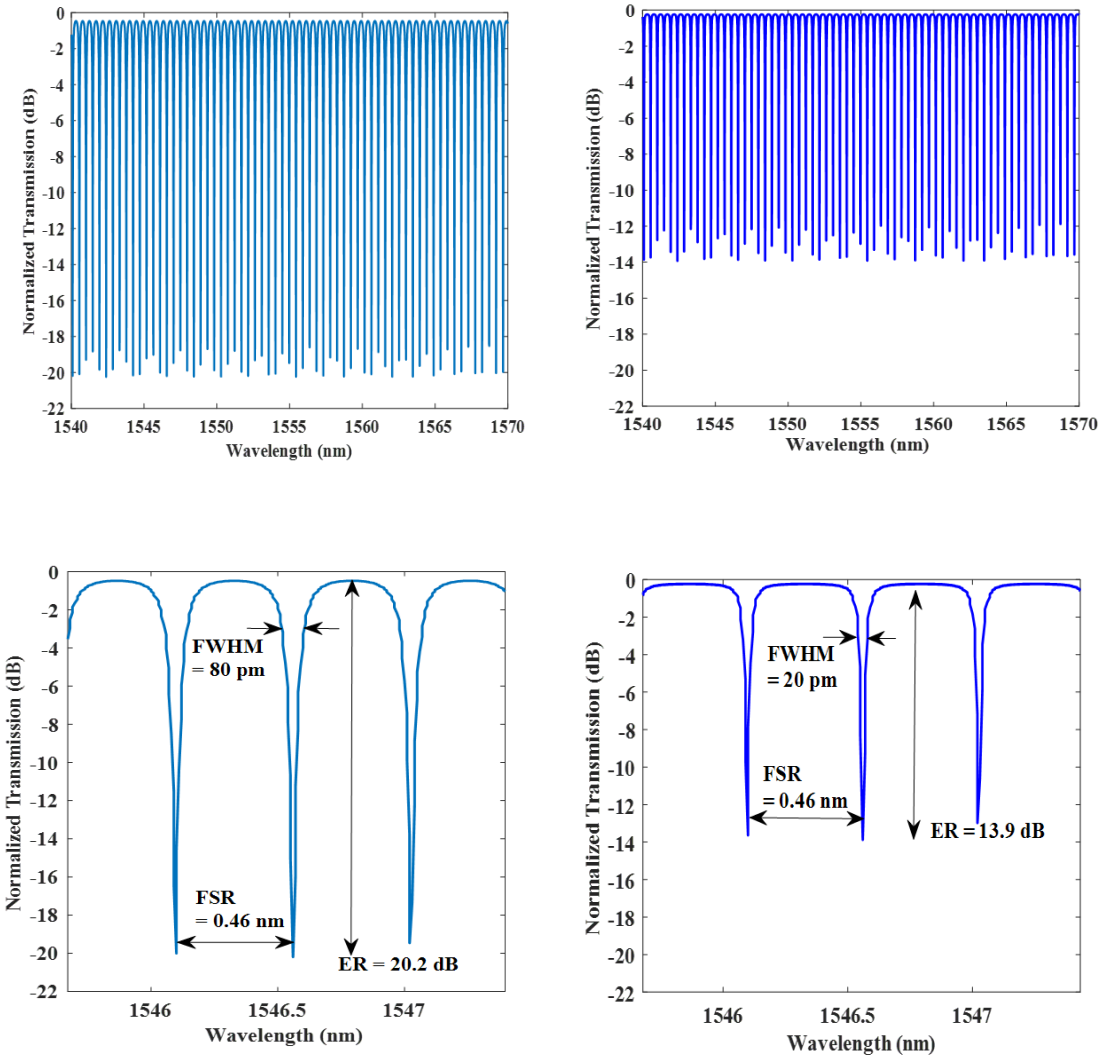



(a) (b)

**Figure 3.3:** Dispersion coefficients of a 400 nm thick SiN waveguide with a bend radius of 400 µm as a function of wavelength. (a) Comparison of dispersion profiles for waveguide widths ranging from 0.5 µm to 2.5 µm. (b) Zoomed-in view of the dispersion for the 2 µm wide waveguide.

The coupling gap ($g$) between the bus and ring waveguide is optimized through 3D FDTD and is set to 220 nm. The coupling length is also optimized to 30 µm using FDTD, set to achieve a high extinction ratio at resonating wavelengths. The transmission response of the simulated 400 µm racetrack SiN microring resonator is done using the circuit level simulator module Interconnect by Ansys Lumerical by importing the mode properties of both the straight and ring waveguides from the FDE simulation results and the coupling coefficient obtained through FDTD simulation. The SiN propagation losses for the ring and straight waveguides are included in the circuit-level simulation.

The specific propagation loss of a 1 µm wide waveguide is not explicitly known for our fabricated SiN die. However, the die manufacturer, ANT, reports propagation loss data for a 750 nm-wide straight waveguide (1.85 dB/cm) and curved waveguide (10.6 dB/cm). ANT confirmed by email (Jocelyn Bachman, personal communication, 23 January 2025) that a 1 µm wide waveguide generally exhibits 0.1 dB/cm lower propagation loss than the 750 nm wide waveguide. ANT also noted that propagation loss varies across the die and that values below 0.8 dB/cm may not be realistic. Additionally, in a follow-up communication on February 3, 2025, ANT referenced older data from 2022, indicating that some of their SiN fabricated dies exhibited a propagation loss of 1.1 dB/cm for a 1 µm wide straight waveguide. However, they did not provide data on the propagation loss for a 1 µm wide curved waveguide Based on this information, the simulated 400-µm SiN racetrack microring resonator includes the propagation loss of 10.5 dB/cm for the curved waveguides and 1.75 dB/cm for the straight waveguides. A wavelength resolution of 5 pm is used in this simulation across 1540 nm to 1570 nm optical bandwidth. The resulting transmission response over the optical bandwidth range is shown in Fig 3.4(a) exhibits an insertion loss of 0.5 dB and an extinction ratio from 18.6 to 20.2 dB. The maximum extinction ratio obtained is 20.2 dB for the resonance at a wavelength of 1546.56 nm with a FWHM of 80 pm. The FSR is 0.46 nm as shown in the zoomed-in view of Fig 3.4(a). These results lead to a calculated loaded Q factor of approximately 19,332 at 1546.56 nm wavelength.




(a) (b)

**Figure 3.4:** Normalized transmission (top row) and zoomed-in around 1546.56 nm (bottom row) of the simulated SiN 400 µm microring racetrack resonator. Results with a) a propagation loss of 10.5 dB/cm for the ring waveguide and 1.75 dB/cm for the straight waveguide and (b) with a propagation loss of 3.2 dB/cm for the ring and 2.2 dB/cm for the straight waveguide.

In the second simulation scenario, the propagation loss is set to 2.2 dB/cm for 2 µm wide straight waveguides (based on ANT's confirmation of 1.1 dB/cm for 1 µm wide waveguides) in the Interconnect simulation. The ring waveguide is simulated with a propagation loss of 3.2 dB/cm in ANSYS/Lumerical Interconnect to evaluate the impact of loss on performance, as the propagation loss for the curved waveguide remains unknown. The resulting transmission response, shown in Fig 3.4(b), exhibits a maximum extinction ratio of 13.9 dB at 1546.56 nm much lower than in the first scenario, with a narrower FWHM of 20 pm. The zoomed-in view reveals an unchanged FSR of 0.46 nm. These results lead to a calculated loaded Q factor of approximately 77,328, which is much higher than in the first scenario due to the lower loss in the ring waveguide for the resonance at 1546.56 nm wavelength. The insertion loss is lower, 0.24 dB relative to 0.5 dB for the first scenario.

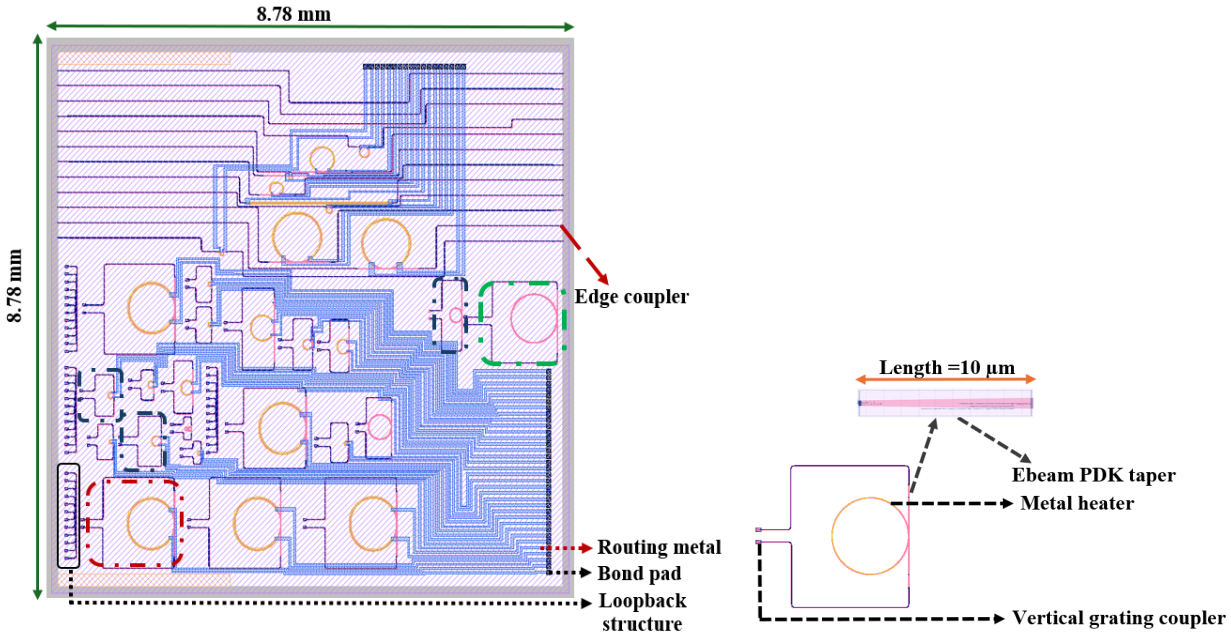
From the transmission response analysis of the two loss assumption scenarios, lower propagation loss in the ring waveguide significantly increases the extinction ratio and Q factor of the resonator, while also leads to lower device insertion loss. The straight waveguide of 800 µm propagation loss of 0.18 dB is the main contributor to the overall loss of the microring resonator. Since the precise propagation loss for the SiN die is not known, a value of 2.2 dB/cm is selected for the bus waveguides and 3.2 dB/cm for the ring waveguide in 400 µm SiN racetrack microring resonator. A summary of the simulated performance of 400 µm racetrack microring is presented in Table 3.1

**Table 3.1:** Summary of Simulated Racetrack Microring Performance

| **Ring and Straight Waveguide Width (µm)** | **Ring Waveguide Propagation Loss (dB/cm)** | **Straight Waveguide Propagation Loss (dB/cm)** | **FSR (nm)** | **FWHM (pm)** | **ER (dB)** | **Calculated Q Factor** |
|---|---|---|---|---|---|---|
| 2 | 10.5 | 1.75 | 0.46 | 80 | 20.2 | 19332 |
| 2 | 3.2 | 2.2 | 0.46 | 20 | 13.9 | 77328 |

## 3.2 SiN Design Layout

The design layout of the SiN microring resonator is done using Klayout tools. The GDS file was sent for a subsidized fabrication at Applied Nanotools Inc (ANT) through CMC Microsystems, in April 2022. The layout of the die area space provided of 8.78 mm × 8.78 mm is shown in Fig. 3.5(a) which includes several replicas of the main ring design with variations in waveguide width, ring radius, coupling gap, and coupling length based on previous FDE optimizations, with ranges spanning: 0.5–2.5 µm for width, 30–400 µm for radius, 200–500 µm for coupling gap, and 0–70 µm for coupling length to explore the impact of these variations on desired performance.




(a) (b)

**Figure 3.5:** (a) Chip layout with various SiN microring resonators and test structures. The red dotted rectangle indicates the measured 400 µm SiN racetrack microring, while the green dotted rectangle highlights the measured 400 µm SiN circular microring; the dark blue dotted rectangles indicate the other measured three microrings with radii of 50 µm, 83 µm and 100 µm; (b) zoomed-in view of the red rectangle labeled microring resonator with vertical grating couplers.

The design layout shown in Fig. 3.5(b) focuses on the 400 µm racetrack microring with a 30 µm straight coupling region, as described in section 3.1. Among all the variations, five microring resonators are evaluated (indicated by the dotted rectangle in the layout), including two 400 µm radius racetrack resonators (marked with a red dotted rectangle) and a circular microring (marked with a green dotted rectangle). The geometrical parameters of

the measured design variations are summarized in Table 3.2

**Table 3.2:** Design Geometries of Measured Circular and Racetrack Microrings

| Ring Radius (µm) | Waveguide Width (µm) | Gap (nm) | Coupling Length (µm) |
|---|---|---|---|
| 50 | 1.5 | 350 | 0 |
| 83 | 1.3 | 430 | 0 |
| 100 | 2.0 | 500 | 0 |
| 400 | 2.0 | 220 | 30 |
| 400 | 2.0 | 220 | 0 |

Of these five microrings in Table 3.2, some devices exhibited resonance wavelengths in the S-band ($< 1530$ nm) or the L-band ($> 1565$ nm), which made it unfeasible to measure their performance using vertical grating couplers that peak in the C-band. The first circular microring variation with a 50 µm radius, a 1.5 µm waveguide width, and 350 nm coupling gap exhibits a calculated loaded Q factor as high as 84,288 at 1517.59 nm, but with an FSR of 3.6 nm which is reduced the generated number of comb lines. The second circular microring device with a 83 µm ring radius and a 0.43 µm coupling gap exhibits a calculated loaded Q-factor of 60,934 at 1523.38 nm, with an FSR of 2.2 nm. The 100 µm radius microring with a 2 µm waveguide width and the largest coupling gap of 500 nm achieves a Q-factor of 56,818 at 1590.9 nm, with an FSR of 1.8 nm. Despite the high Q factors observed in the 50 µm and 83 µm microrings, they are not suitable for OFC generation due to their high normal dispersion characteristics. The 50 µm microring (with a 1.5 µm width) and the 83 µmicroring (with a 1.2 µm width) exhibit significant negative dispersion, ranging from -400 ps/nm · km to -600 ps/nm · km in the 1.5 µm to 1.6 µm wavelength range. Additionally,

their resonances in C-band cannot be selected for two factors: (1) higher normal dispersion in the C-band (<- 423 ps/nm.km for 50 µm microring and <- 524 ps/nm.km for 83 µm) and (2) lower extinction ratios in the measured resonances, which ranges from 5.2 dB to 2.8 dB for the 50 µm microring and 7 dB to 3.1 dB for the 83.3 µm microring. Additionally, the microrings exhibit relatively large FSR values (3.6 nm, 2.2 nm), which adversely affect FWM interactions and reduce the number of generated comb lines within a specific bandwidth. A larger FSR decreases the density of resonant modes within the specific bandwidth and increases the mode spacing, thereby limiting spectral overlap required for efficient nonlinear interactions. As a result, when pump waves interact in FWM process to generate sidebands, the reduced mode availability in the microring weakens nonlinear coupling, leading to lower FWM efficiency [146].

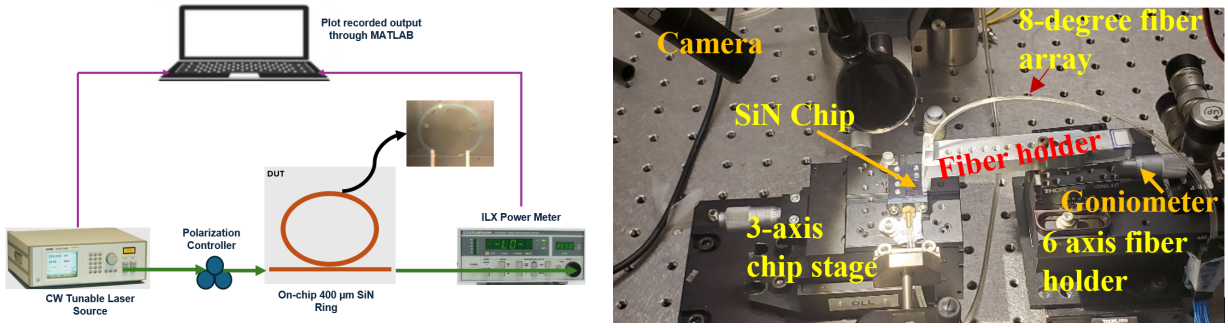
The selected two functional 400 µm microrings, the racetrack with a 30 µm coupling length and the circular one with no coupling length, have resonances within the C-band (1530–1565 nm). The measured results of these two resonators are analyzed next in this section and section 3.4.

The metal layer of titanium-tungsten (TiW) alloy is used above the rings as a heater in the event that the ring resonating wavelength needed to be tuned. The metal heater is connected to two bond pads via the routing layer, as shown in the layout of Fig. 3.5. The resonators are interfaced via vertical grating couplers that are designed for operation in the TE mode. Vertical grating coupler loopbacks are included to measure the fiber-to-die

coupling loss. Versions of the microring interface via edge couplers for an off-chip interface for lower coupling loss and maintaining a broader optical bandwidth [147]. However, these microrings were not measured due to alignment limitations in the testbed. The design layout of both these couplers is from ANT's Process Design Kit (PDK). The couplers are connected through tapers (also from ANT's PDK) to match the bus waveguide's width with the width of the grating couplers.

## 3.3 SiN Racetrack Microring Resonator

The fabricated SiN die is experimentally measured at McGill University, in our research team laboratories in the McConnell Engineering building. The experimental setup use to characterize the resonance of the microrings is illustrated in Fig. 3.6.




(a) (b)

**Figure 3.6:** a) Schematic of the experimental test setup used to measure the fabricated SiN circular and racetrack microring (photo image shows the racetrack ring from the fabricated die) (DUT: device under test, CW: continuous wave). b) Photography of the experimental laboratory setup with labeled key testbed components.

The experimental validation is focused on the optical transmission response of the various microrings measured. I used a tunable C-band laser (Model Tunics T100R, manufactured by Yenista) operating over a wavelength range from 1490 to 1600 nm, with a minimum scanning resolution set to 10 pm, which is sufficient to determine the Q factor. Although a higher-resolution laser of 1 pm can be employed for more precise measurements. However, increasing the wavelength resolution does not affect the Q factor and remains on the order of $10^4$. The Q factor is primarily limited by the propagation loss in the ring waveguide, as supported by the simulation results presented in Section 3.1. Since the desired resonance wavelengths and corresponding Q factor are determined at a 10 pm resolution, higher resolution was not necessary for this analysis. The continuous-wave (CW) optical output power of the laser is 0 dBm. The optical signal first passes through a polarization controller to align the polarization state to the on-chip transverse electrical (TE) mode. The light is then coupled in and out the SiN die (DUT) through the vertical grating couplers via an 8-degree fiber array of 12 channels (only channels 1 and 2 are used). The fiber array is fixed onto a 6-axis fiber holder stage with a goniometer to optimize the fiber array angle with respect to the die. The die is on a 3-axis chip stage (Fig. 3.6(b)). The optical experimental power is measured using an optical power meter (model FPM-8220, manufactured by ILX), which has a dynamic range from -75 dBm to +1.5 dBm. The optical transmission response is recorded using a MATLAB-scripted interface through GPIB interface cards between the laser and the power meter.

The fiber-to-chip coupling loss is measured using the loopback structure labeled in the layout (Fig. 3.5(a)) which includes two vertical grating couplers. In Fig. 3.7, the measured optical transmission corresponds to the optical power measured by the ILX optical power meter with the optical input power at 0 dBm. As labeled, the maximum transmission occurs at 1597.0 nm with a measured optical power of -12.47 dBm, corresponding to 6.2 dB optical power loss per vertical grating coupler, which means that more than 90% of the optical power is lost solely in coupling light in and out of the die. This illustrates the challenges of performing the optical nonlinearity required to generate an OFC laser.

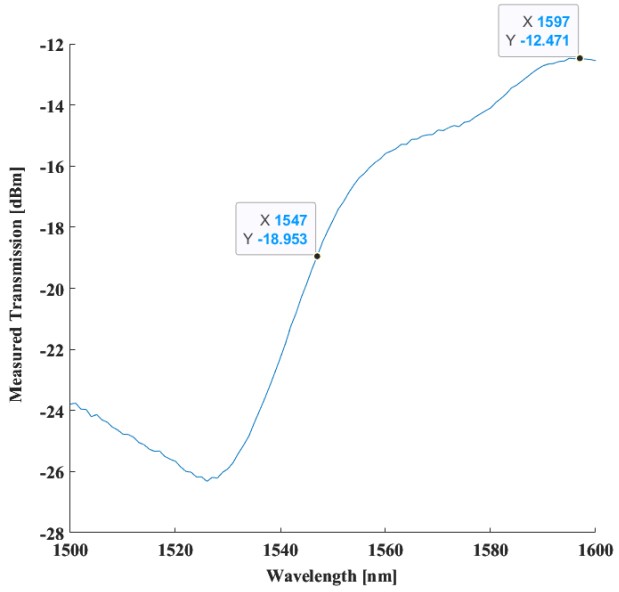
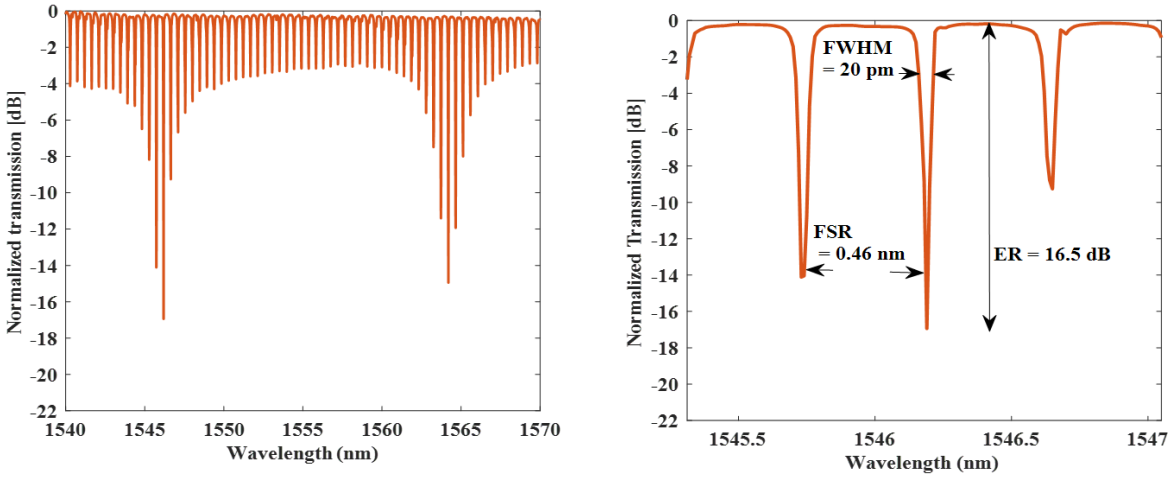



**Figure 3.7:** Measured optical transmission of the loopback structure with two SiN vertical grating couplers. The optical input power is 0 dBm.

The expected peak transmission for the vertical coupler designs used from ANT's PDK is expected to be a shorter wavelength, 1547.0 nm, and not 1597.0 nm. Further, the expected

loss is 5.5 dB per coupler at that wavelength. The fabrication process likely exhibited an over-etching processing step that led to a red shift of the transmission peak from the intended operating wavelength of 1547 nm. At 1547 nm, the measured loss corresponds to 9.5 dB per grating coupler, 4 dB more than expected. This loss corresponds now to close to 99% of the light lost in the coupling process, leaving 1% of the original light in the ring structure. The hope of confirming the simulation results vanishes at this point, and the experimental focus is on confirming the ring performance alone. I continued to operate at 1547 nm as the dispersion is too high at longer wavelengths, further reducing the nonlinear efficiency of the microring resonator.

The experimentally measured transmission response of the SiN racetrack microring is depicted in Fig. 3.8, normalized to the spectral response of the loopback test structure (Fig. 3.7). The extinction ratio ranges from 3.8 dB to 16.5 dB due to the low scanning resolution wavelength of 10 pm. We select the resonance at 1546.19 nm which has an maximum extinction ratio of 16.5 dB. A closer view of the measured transmission response is shown in Fig. 3.8(b) where the resonance has a an FWHM of 20 pm where we have obtained 17 pm through simulation. With the FSR of 0.46 nm aligned with our simulation results, the calculated loaded Q factor is 77,800 at 1546.19 nm. The measured insertion loss of the ring resonator is 0.3 dB at 1546.19 nm.

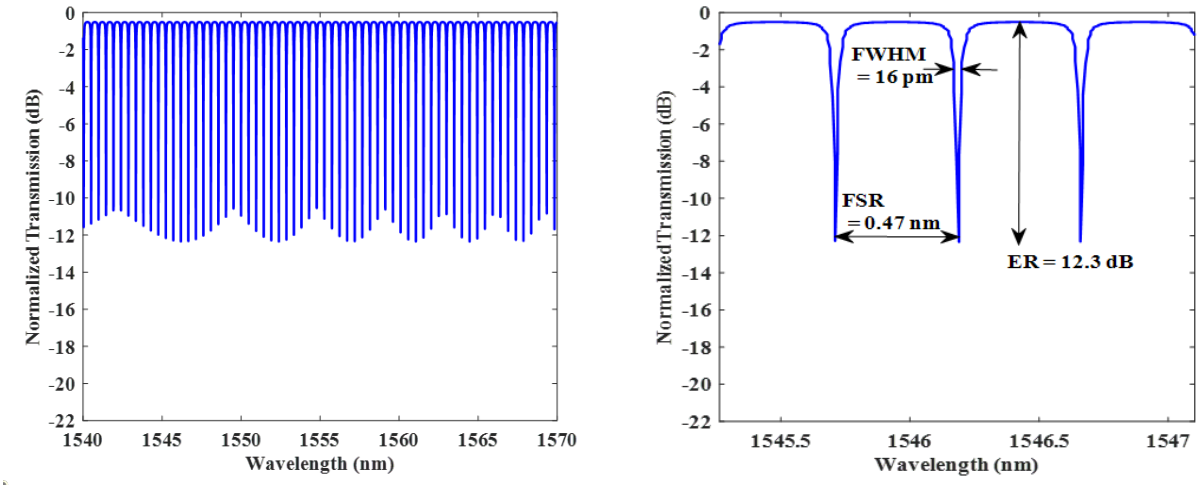



(a) (b)

**Figure 3.8:** (a) Measured transmission response of the 400 µm SiN racetrack resonator, (b) zoomed-in view at 1546.19 nm resonating wavelength.

## 3.4 SiN Circular Microring Resonator

The geometry of the circular microring, with a radius of 400 µm, is identical to the racetrack microring in Fig. 3.1 except for the ring-to-waveguide coupling region. In the circular microring, the coupling region has a 220 nm coupling gap but no defined coupling length. The transmission response of the 400 µm circular microring obtain through simulation with a propagation loss of 3.2 dB/cm for the ring waveguide and 2.2 dB/cm for the bus waveguide is shown in Fig 3.9(a).

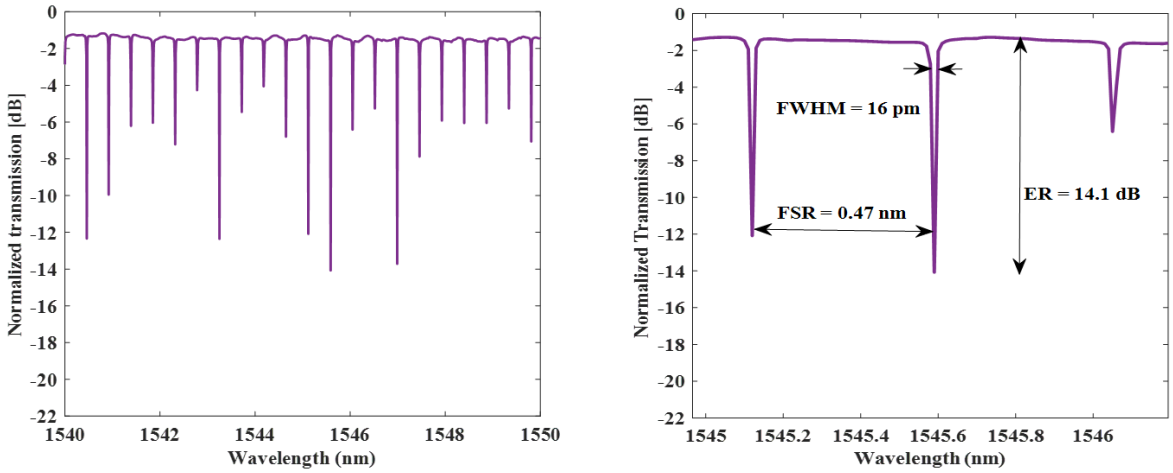



(a) (b)

**Figure 3.9:** (a) Normalized transmission response obtained through simulation with a propagation loss of 3.2 dB/cm for the ring and 2.2 dB/cm for the bus waveguide, (b) zoomed-in view around 1546.2 nm wavelength

The transmission exhibits a maximum extinction ratio of 12.3 dB at 1546.2 nm with an insertion loss of 0.55 dB. The zoomed-in view in Fig 3.9(b) reveals an FSR of 0.47 nm and FWHM of 16 pm at 1546.2 nm. The corresponding calculated loaded Q factor is 96,637.

The measured circulator ring resonator exhibits a higher insertion loss of 1.5 dB at 1545.59 nm wavelength, as shown in Fig. 3.10(a). The measured transmission response reveals extinction ratios ranging from 4.26 dB to 14.1 dB. At 1545.49 nm, the largest extinction ratio of 14.1 dB is obtained with an FWHM of 16 pm (Fig. 3.10(b)). The FSR is 0.47 nm. The calculated loaded Q factor is 96,599 at 1545.59 nm.




(a) (b)

**Figure 3.10:** a) Measured normalized transmission spectrum of the 400 µm circular microring with b) a zoomed-in view of resonance at 1545.59 nm wavelength.

**Table 3.3:** Experimental performance comparison of the SiN racetack and circulator microring resonators.

| **Type** | **Radius (µm)** | **Coupling Length (µm)** | **FSR (nm)** | **ER (dB)** | **FWHM (pm)** | **Calculated loaded Q-factor** | **Insertion loss (dB)** |
|---|---|---|---|---|---|---|---|
| Racetrack Resonator | 400 | 30 | 0.46 | 16.5 | 20 | 77,800 | 0.3 |
| Circular Resonator | 400 | 0 | 0.47 | 14.1 | 16 | 96,599 | 1.5 |

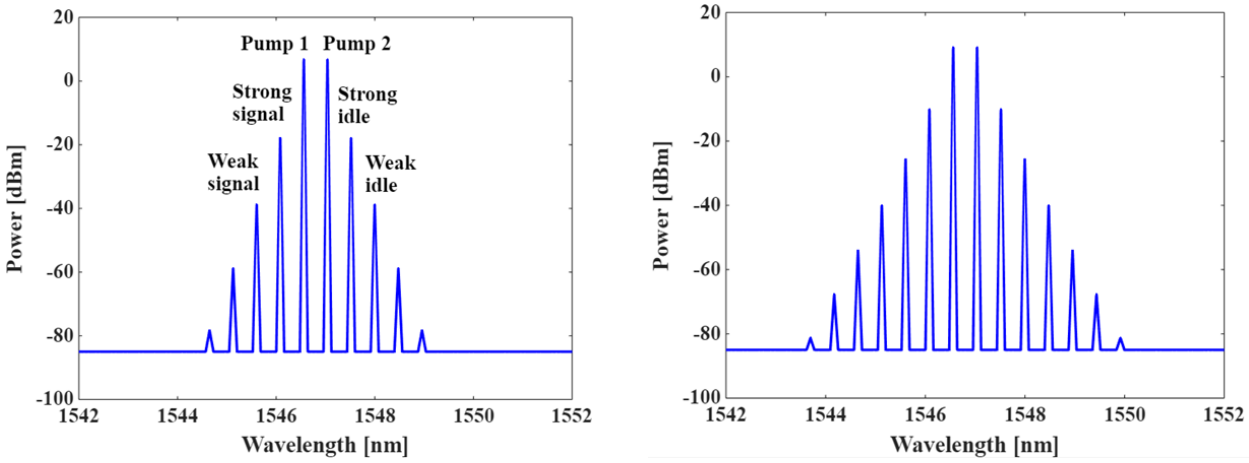
The simulation results show that the insertion loss of the circular microring is higher, at 0.55 dB, compared to the racetrack microring, which has an insertion loss of 0.33 dB. This difference is attributed to the sharper bends in the coupling region of the circular microring. Additionally, Table 3.3 compares the measured performance of the racetrack resonator and

the circular resonator. The racetrack resonator provides a larger FWHM of 20 pm relative to 16 pm for the circulator resonator since the racetrack resonator has a higher optical path length, as it incorporates two straight waveguides in addition to two semi-circular bends. Consequently, the calculated loaded Q factor is greater and thus better for the circular resonator considering that they have similar FSR. However, the circular resonator exhibits experimentally higher insertion loss (1.5 dB) compared to the racetrack resonator (0.3 dB). Incorporating a 30 µm coupling length in the microring racetrack resonator leads to lower insertion loss likely due to the geometry of the design. In a circular resonator, the bending loss is constant along its entire optical path due to its continuous and uniform curvature. In contrast, a racetrack resonator consists of two straight waveguides and two semi-circular bends. As a result, bending loss occurs only in the semi-circular sections, while the straight waveguides experience negligible bending loss. The racetrack resonator exhibits higher coupling efficiency since its straight waveguide couple more effectively with the bus waveguide in the coupling region. This coupling region enhances the propagation of light in the microring. Therefore, the racetrack configuration minimizes losses associated with sharp bends, effectively reducing the overall insertion loss [148]. Therefore, the performance of the designed 400 µm SiN microring racetrack resonator is slightly more suitable for low-loss and nonlinear optics applications.

## 3.5 Optical Frequency Comb Simulation

An analytical approach is employed to generate the OFC using simulation characterized results for the 400 µm SiN microring racetrack resonator using the nonlinear Schrodinger equation (NLSE) with non-degenerate four-wave mixing (FWM). The numerical simulation uses the split-step Fourier method (SSFM) to solve the NLSE (Eq. 2.13). The microring is modeled as a closed-loop system, with the intracavity field evolving over 30 round trips. At each round trip, the NLSE (Eq. 2.13) governs the propagation of two pump fields at distinct wavelengths, and their interaction via FWM leads to the gradual buildup of new spectral lines. The SSFM discretizes each round trip into small propagation steps, updating the field to capture nonlinear coefficients, nonlinear phase shifts, group velocity dispersion, and round-trip loss to effectively modeling comb generation dynamics within the cavity. The extinction ratio of 13.9 dB measured at 1546.56 nm wavelength requires the input pump sources to be optimally tuned to the resonance wavelength to enhance FWM efficiency. The group velocity dispersion is calculated using equation 2.14, incorporating the simulated waveguide dispersion values. The simulation considers the SiN microring racetrack resonator's characterized parameters, such as dispersion, refractive index, FSR, calculated Q-factor, insertion loss, and FWHM. The nonlinear refractive index of SiN material used is $2.4 \times 10^{-19}\,\mathrm{m}^2/\mathrm{W}$ [149] in equation 2.17 to determine the nonlinear parameter ($\gamma$) of 1.35 rad/W/km for analytical OFC simulation. Non-degenerate FWM, based on equation 2.19, is used to simulate OFC generation with equal input powers of

14.3 dBm to overcome optical losses of 0.24 dB and achieve equally spaced comb lines of 0.48 nm. This simulation excludes fiber-to-chip coupling loss to focus solely on determining the minimum power needed within the microring to facilitate FWM and generate the OFC.




(a) (b)

**Figure 3.11:** Simulated results of the OFC spectrum in a 400 µm radius SiN microring racetrack resonator for pump wavelengths at 1546.56 nm and 1547.02 nm. (a) Two equal-power pumps with input power of 27 mW (14.3 dBm). (b) Two equal-power pumps with input power of 60 mW (17.8 dBm).

In the simulated OFC generation, two pumps with equal minimum power of 14.3 dBm (27 mW) at 1546.56 nm (pump 1) and 1547.02 nm (pump 2) wavelengths are sufficient to excite the sidebands within the C-band. The outermost sidebands drop below -30 dBm at lower optical power due to large normal dispersion. Normal dispersion reduces FWM efficiency, leading to weaker new sidebands [83]. The OFC spectrum in Fig. 3.11(a) shows six comb lines (up to -38 dBm optical power), including the two pump sources and the two

smaller peaks on the right and left representing the idler and signal fields. The strongest idler and signal resonances occur at 1547.52 nm and 1546.08 nm, respectively. Conversely, the weakest idler and signal resonances are at 1548 nm and 1545.6 nm, respectively. The spacing between two adjacent comb lines is 0.48 nm. By increasing the input pump powers to 17.8 dBm (60 mW) each, the optical power of the weakest signal and idler is improved up to -22 dBm. This increase in pump power enhances FWM efficiency in the normal dispersion regime, resulting in the generation of fourteen comb lines, of which six are usable.

This OFC generation could not be experimentally validated due to the high fiber-to-chip coupling loss of approximately 9.5 dB per vertical grating coupler at a wavelength of 1547.19 nm, reducing the two pump powers by 8.3 dB, insufficient for OFC generation through FWM. I attempted to measure a microring resonator using edge couplers to minimize fiber-to-chip coupling loss. However, due to alignment challenges and time constraints in gaining the necessary expertise, I was unable to achieve a coupling loss lower than 16 dB per facet. A brief discussion is provided on the challenges associated with both grating and edge couplers, along with potential solutions to mitigate coupling losses in Chapter 5. Additionally, a proposed experimental approach to generate OFC is presented in the future work section in Chapter 5 (Section 5.2).

## 3.6 Summary

The chapter represented the design, simulation, and experimental characterization of a 400 µm SiN microring racetrack resonator with a 30 µm coupling length. The geometry of the design was optimized through simulation to obtain a normal dispersion of -359 ps/nm.km at 1547 nm wavelength. Experimental characterization of the microring reveals a maximum extinction ratio of 16.5 dB for resonance at 1546.19 nm with an FWHM of 20 pm, and an FSR of 0.46 nm. Thus, this microring resonator exhibits a loaded quality factor ($Q_L$) of 77,800. This Q-factor value is primarily limited due to high propagation loss in the ring waveguide. The measured insertion loss is 0.3 dB. A detailed comparison of the measured parameters for two 400 µm microrings, racetrack resonator (with 30 µm coupling length) and circular resonator (with no coupling length), validates that the racetrack resonator provides lower insertion loss of 0.3 dB than the circular microring, and thus is more suitable for OFC generation. These results align with recent research indicating that racetrack-structured SiN microring resonators exhibit reduced insertion loss compared to conventional circular microring resonators. This performance advantage arises from the racetrack geometry, which mitigates radiation losses commonly associated with curved waveguides. In circular microring resonators, continuous bending leads to significant bending losses, which degrades the overall efficiency. Conversely, racetrack resonators incorporate straight waveguide sections that enhance light coupling to the ring, enabling more efficient coupling and lowering propagation losses [150, 151]. This

design mitigates losses and enhances the Q-factor, essential for enhancing performance in applications such as optical frequency comb generation [151, 152].

Using the simulated characterized parameters of the 400 µm racetrack microring in my OFC's numerical simulation, the racetrack resonator demonstrates the potential to generate six comb lines through non-degenerate FWM, including two pump waves with equal optical power of 17.8 dBm (60 mW) at 1546.56 nm and 1547.02 nm. The generated adjacent comb lines are spaced 0.48 nm apart, which differs slightly from the FSR of the microring due to significant normal dispersion.

# Chapter 4

# Actuator-based Silicon Nitride Phase Modulator

Integrated devices based in silicon nitride (SiN) technology are highly appealing for numerous applications due to SiN ultra-low propagation loss characteristics. A SiN phase modulator can enhance the functionality of optical gyroscopes leading to improved rotational sensitivity and compactness relative to current bulky modulators. This chapter introduces a design methodology through simulation of a double lead zirconate titanate (PZT) actuators on a SiN waveguide towards developing a low voltage phase modulator. The double PZT actuator design improves over the single PZT version by proposing a new design to reduce the optical loss in the waveguide structure inherent to the single PZT design. The double PZT structure allows the actuation to be closer to the waveguide

material leading to a more efficient sensing. The novel double PZT-SiN configuration is achieved by optimizing the dimensions of the PZT structures, SiN core waveguide width, and cladding thicknesses. An analysis of a simulated stress distribution across different designs and corresponding changes in the effective refractive index identifies the best layout to achieve the maximum phase shift at low voltage. The following section focuses on the design geometries. The next section discusses the concept and process related to stress distribution in PZT actuators on SiN waveguides, providing a clear understanding supported by simulated results.

## 4.1 Design Specifications

A cross-sectional view of a double PZT actuator on a single-stripe SiN waveguide is depicted in Fig. 4.1. The SiN core with 2.8 µm in width and 100 nm in thickness is placed on top of a 8 µm thick box oxide layer and enclosed by a 8 µm thick $SiO_2$ cladding. The dimension of the SiN core waveguide is recommended by OSCP to ensure single-TE mode operation at this specific core thickness. The design features two PZT actuators, each consisting of a PZT layer with a thickness of 1.5 µm sandwiched vertically between two metal layers that serve as electrodes. The top and bottom electrodes are composed of platinum (Pt) layers with optimized thicknesses of 300 nm and 100 nm, respectively [154]. These values were used as reference points in a COMSOL Multiphysics parameter sweep, where a range of smaller and larger thicknesses was explored. The analysis confirmed that the reference values of

300 nm (top) and 100 nm (bottom) offered the most favorable performance in the design. These PZT actuators are placed above and laterally adjacent to the SiN waveguide such that the mechanical stress is induced from the PZT actuators by applying voltage in our case, onto the SiN waveguide structure. Initially, the distance between the two PZT structures is 22.8 µm and each PZT structure has a width of 25 µm. This design follows a commercial SiN fabrication process stack. Specifically, it can be fabricated using the layer arrangement developed by LioniX [132].

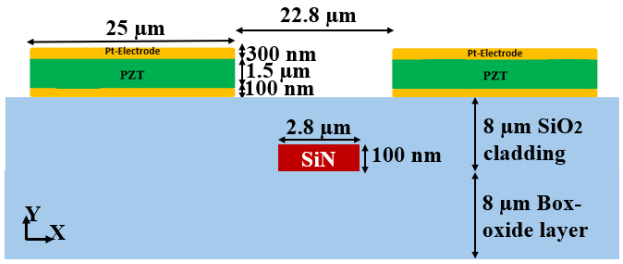



**Figure 4.1:** Cross-sectional view of the double PZT on a SiN waveguide for an actuator-based stress-optic phase modulator.

## 4.2 Design of the Single PZT Actuator-based SiN Phase Modulator

The SiN phase modulator is first simulated with one PZT actuator placed right above the SiN waveguide as the baseline in the performance comparison with our proposed customized double PZT actuator-based SiN phase modulator. The simulated y-stress

tensor for the single PZT structure on a single-strip SiN waveguide is shown in Fig. 4.2, for an applied voltage of 40 V to the bottom electrode with the top electrode grounded (0 V). The piezoelectric coefficient $e_{33}$ of 23.24 C/m$^2$ is used in the simulation. As discussed in chapter 2, the piezoelectric coefficient is the material's response to a voltage applied along this y-direction. Mechanical properties are characteristics of a material that determine the material's mechanical strength and ability to be shaped or formed. The important mechanical property of the materials in this investigation is the Young's modulus, a material stiffness measurement under compression, which is 70, 95, 168, and 270 GPa for $SiO_2$, PZT, Pt, SiN, respectively [131].

According to the inverse piezoelectric effect in the PZT structure, applying a voltage of 40 V to the bottom electrode with the top electrode grounded generates an electric field between the two electrodes that deforms in the PZT layer, acting as the actuator. This deformation generates mechanical stress that can be quantified using the piezoelectric constitutive equations (2.23) and (2.24). The induced mechanical stress propagates below in the silica cladding and eventually through the SiN waveguide, altering its refractive index. Fig. 4.2 illustrates the decompression (red regions) at the edge of the PZT structure, but more important to this work, we observe a compressive stress (blue regions) of approximately -3.3 $\times$ 10$^7$ N/m$^2$ reaching the SiN waveguide.

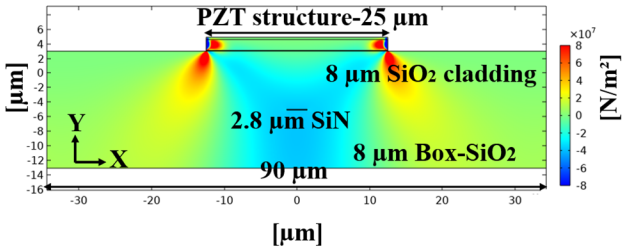
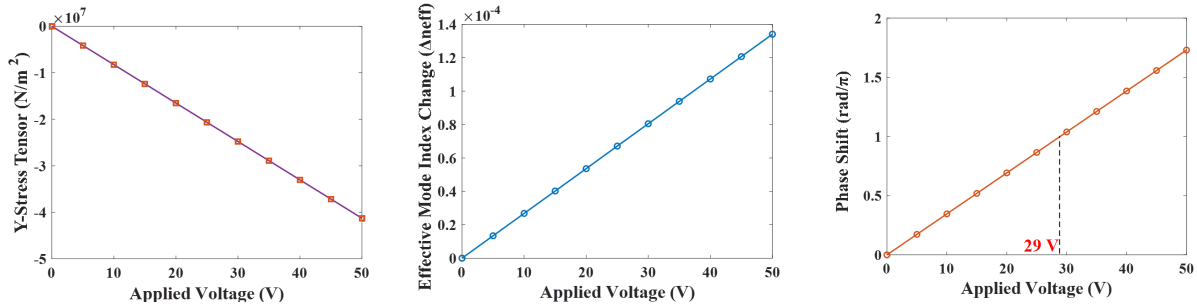



**Figure 4.2:** Stress tensor obtained through simulation in the xy-plane for a single PZT actuator with an applied voltage of 40 V on the bottom electrode and 0 V on the top electrode. The blue shading represents compressive stress (negative stress) which reaches the SiN waveguide, and the red-to-yellow shading indicates tensile stress or decompression (positive stress).

The applied voltage on the bottom electrode of the PZT actuator varies from 0 to 50 V, while the top electrode is maintained at 0 V (grounded). This voltage variation is used to evaluate the stress distribution in the SiN waveguide. The plot of the y-stress tensor as a function of applied voltage, shown in Fig 4.3(a), indicates that compressive stress (negative stress) increases with increasing voltage.

Using the Finite Element Method (FEM) for electromagnetic wave optics simulation in COMSOL Multiphysics, the effective refractive index of the SiN waveguide is calculated for various stress range by varying the applied voltages ranging from 0 to 50 V. This stress significantly changes the refractive index of the SiN waveguide's material through the stress-optic effect. The refractive indices of SiN and $SiO_2$ are 1.9835 and 1.455, respectively, at a wavelength of 1550 nm. The refractive index changes ($\Delta n$) in SiN waveguide along the $x$, $y$, and $z$ direction is found based on the respective spatial stress tensor components as

defined by Eqs. 2.25, 2.26, and 2.27, in Chapter 2. We obtain the refractive index changes ($\Delta n$) in directions using stress tensor components and stress-optic coefficients $C_1$ and $C_2$ of $-4 \times 10^{-13}$ 1/Pa and $2.4 \times 10^{-11}$ 1/Pa that are derived from Eqs. 2.28 and 2.29.




(a) (b) (c)

**Figure 4.3:** (a) Y-stress tensor distribution in the SiN waveguide as a function of the applied voltage across the full width of the PZT structure.(b) effective refractive index change of the optical fundamental mode in the SiN waveguide as a function of the applied voltage on the PZT structure. (c) corresponding phase shift as a function of the applied voltage on the PZT actuator and the y-direction compression stress.

The stress-optic effect expression determines the refractive index changes in the SiN waveguide and is used in COMSOL Multiphysics to obtain the effective mode index in the waveguide through simulation. Since the stress-optic coefficient $C_2$ is much larger than $C_1$ and considering only the light propagating in the TE mode, the maximum refractive index change is obtained in the x-direction ($\Delta n_x$) caused by the compressed stress in the y-direction ($\sigma_y$). Fig. 4.3(b) illustrates the relationship between the effective index change ($\Delta n_{eff}$) and the applied voltage. The effective index change increases with increasing voltage, as expected through Eq. 2.31 in Chapter 2. There is a positive change in the effective index

due to the compressed stress at the waveguide. Consequently, with compressive stress on the waveguide, the phase shift of the propagating light in the waveguide structure increases, as described by Eq. 2.30. Fig. 4.3(c) shows the resulting phase shift as a function of stress. The obtained half-wave voltage $V_\pi$ is 29 V for a 1 cm long modulator design.

The single PZT actuator-based SiN phase modulator configuration in this work is similar to the design described in a 2024 granted patent by Lionix International [154], where they propose an asymmetric double-strip SiN core waveguide with a 2 µm PZT layer. For a simpler structure potentially lowering the fabrication cost, our design employs a single-strip SiN waveguide with a 1.5 µm PZT layer. The single PZT configuration incorporates metal electrodes directly above the SiN waveguide. The interaction between the evanescent optical field and the metal layer introduces additional propagation losses due to metal absorption. In the past, reducing the top cladding thickness from 4 µm to 2 µm increases the metal absorption loss from $10^{-5}$ to close to 1 dB/cm, which in turn leads to a lower $V_\pi$ for the modulator [155]. The single PZT actuator-based SiN phase modulator design serves as a reference for developing a customized configuration aimed at reducing optical propagation loss in the SiN waveguide due to metal absorption. As an initial modification, the single PZT actuator is shifted to one side to evaluate the resulting stress on the SiN waveguide structure. However, this configuration generates only minimal stress in the waveguide. To address this minimal stress limitation, a double PZT actuator configuration is adopted. In this design, the PZT actuators are positioned laterally adjacent to the SiN waveguide, with their separation

specifically recommended by OSCP. This innovative arrangement helps reduce the optical loss due to metal absorption [156].

## 4.3 Performance Analysis

To propose a new stress-optic phase modulator and address the optical loss in the single PZT actuator-based SiN modulator, a double PZT actuator-based SiN phase modulator, as shown in Fig 4.1, is investigated. A design methodology is developed in collaboration with the OSCP team to optimize an efficient double PZT actuator-based SiN phase modulator by investigating key geometric parameters. These parameters included the spacing between the PZT actuators, the top thickness of the cladding $SiO_2$, and the width of the waveguide core, to maximize the phase shift at low driving voltage. The initial structure shown in Fig 4.1 served as the reference for optimization, where the distance between PZT actuators of 22.8 µm is recommended by the OSCP team. The process began by varying the distance between the PZT structures, followed by a simultaneous variation of the cladding thickness and the spacing of the PZT actuators. To further reduce the required voltage, variations in core thickness are incorporated into the optimization. This iterative design process, facilitated through discussions with the OSCP team, resulted in the final optimized structure. No published references are used for this methodology, as existing literature typically reports only the final design without detailing the step-by-step optimization process. A 40 V voltage is applied to both bottom electrodes while maintaining 0 V on

both top electrodes, generating an electric field in the PZT layer of each actuator in Fig 4.1. This electric field induces deformation and mechanical stress in both PZT actuators due to the inverse piezoelectric effect. The induced mechanical stress in the PZT actuators propagates to the SiN core waveguide and surrounding $SiO_2$ cladding. Fig. 4.4 shows the stress tensor on the y-axis ($\sigma_y$) in the simulated double PZT structure on a single stripe SiN waveguide design. This design is referred to as 'Design 1' with reduced metallic absorption with metal layers above the waveguide offset 8 µm laterally. The stress expands along the y-direction due to the effect of tensile stress and undergoes compression along the x-direction. Meanwhile, the waveguide's cladding experiences both compression and tensile stress, while the SiN core is subjected only to tensile stress.

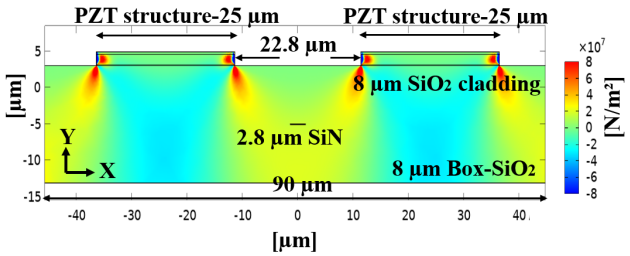
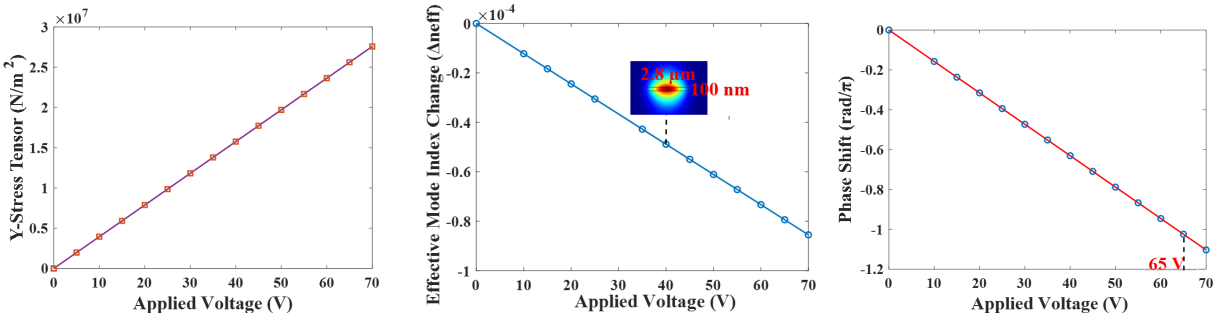



**Figure 4.4:** Resulting y-stress tensor through simulation stress induced from the PZT structure at an applied voltage of 40 V. The blue shading represents compressive stress (negative stress) and red shading represents tensile stress (positive stress).

The applied voltage on both bottom electrodes ranges from 0 to 70 V, while the top electrodes are grounded in the PZT actuators to evaluate the stress distribution in the SiN waveguide. The plot of the y-stress tensor at different voltages, as shown in Fig 4.5(a), indicates that the tensile stress increases as the applied voltage increases. The resulting

stress ($\sigma_y$) in the SiN waveguide is $1.6 \times 10^7\,\mathrm{N/m^2}$ at 40 V.




(a) (b) (c)

**Figure 4.5:** (a) Y-stress tensor in the SiN waveguide as a function of the applied voltage on the double PZT structures in Design 1. (b) effective index change as a function of applied voltage on PZT actuators (single mode profile shown in the inset) (c) corresponding phase in the waveguide over different applied voltage.

Using FEM for electromagnetic wave optics simulation in COMSOL Multiphysics, the effective mode index in the SiN waveguide is determined under varying applied voltage from 0 to 70 V on the PZT structure. The single-mode behavior is maintained across the voltage range while the effective index changes. The TE mode confinement remains high within the core of the SiN waveguide. The decrease in the effective index is attributed to the positive stress induced in the core SiN waveguide causing expansion of the core in the y-direction (Eq. 2.25) as shown in Fig. 4.5(b). The phase shift of propagating light becomes increasingly negative as the applied voltage increases. The half-wave voltage $V_\pi$ is achieved at 65 V for a 1 cm long waveguide in Design 1 as shown in Fig. 4.5(c).

To design an efficient SiN phase modulator, the geometry of Design 1 is varied to achieve an optimized design. Key parameters such as the distance between the two PZT structures, the top cladding thickness, and the core waveguide width are investigated to maximize the phase shift at lower voltage. The results are presented in the following subsections.

### 4.3.1 PZT Separation Optimization

The variation in the waveguide stress is studied as a function of the distance between the two PZT actuators. The phase shift obtained is compared to identify the design that achieves the maximum phase shift at low voltage.

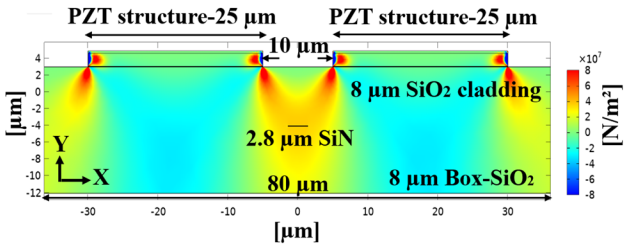
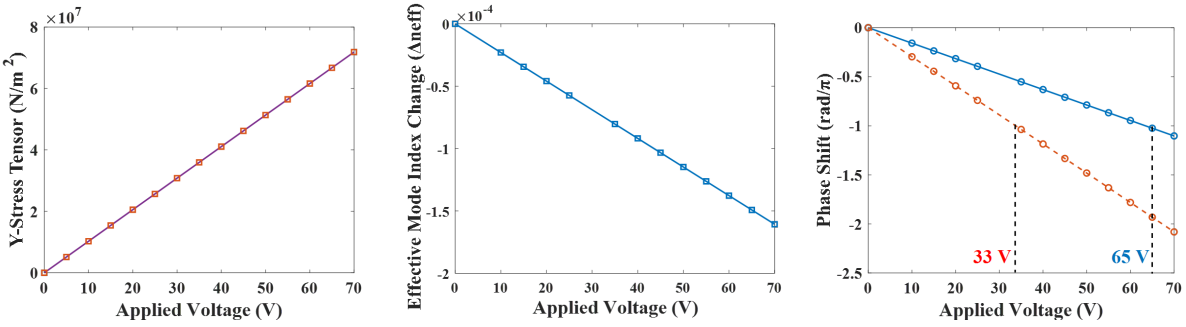



**Figure 4.6:** Simulated y-stress tensor induced in the SiN waveguide with a 10 µm distance between the two PZT structures at an applied voltage of 40 V on the bottom electrode and 0 V on the top electrode. The blue shade represents negative (compressed) stress and the redish-yellow shade represents positive (tensile) stress.

The distance between the two PZT structures was reduced from 22.8 µm to 10 µm. All other designing parameters have remained the same as in Design 1 (Fig. 4.4) where the width of the full design is reduced from 100 µm to 80 µm through simulation. The stress distribution for a 10 µm distance between two PZT actuator-based configurations is shown

in Fig. 4.6 and referred as Design 2. The tensile stress is more focused on the SiN waveguide, creating a U-shaped stress distribution. This causes expansion in the core waveguide with the y-stress tensor reaching a maximum of $4.2 \times 10^7$ N/m$^2$ at 40 V (Fig 4.7(a)). The resulting positive stress in the SiN waveguide increases with the applied voltage such that the effective mode index decreases from the stress-optic effect (Fig 4.7(b)).




(a) (b) (c)

**Figure 4.7:** (a) Plot of y-stress tensor in the SiN waveguide and (b) Plot of effective index change as functions of applied voltage on the double PZT structures of Design 2. (c) Phase shift change as a function of applied voltages of two PZT separations, 22.8 µm and 10 µm distance.

The phase shifts in the waveguide structure for different distances between the PZT structures are determined based on the change in the effective mode index obtained at different applied voltages (Fig. 4.7(c)). Design 2 performs better with a 10 µm distance between two PZT actuators. The half-wave voltage $V_\pi$ obtained is 33 V for Design 2 and 65 V for Design 1 for a 1 cm long device.

### 4.3.2 Top Cladding Thickness Optimization

To further reduce the $V_\pi$ value, the geometry of Design 2 is modified by simultaneously varying the top $SiO_2$ cladding thickness in the waveguide and the distance between the PZT structures. The top cladding varies from 8 µm to 2 µm, while the distance between the two PZTs varies from 10 µm to 3 µm. Fig. 4.8 illustrates the obtained stress for design with a 3 µm top cladding and a 3 µm separation between two PZT structures. This design is referred to as 'Design 3'.

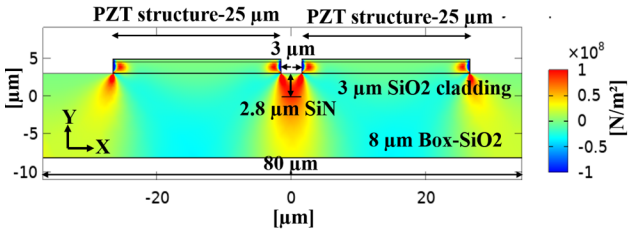
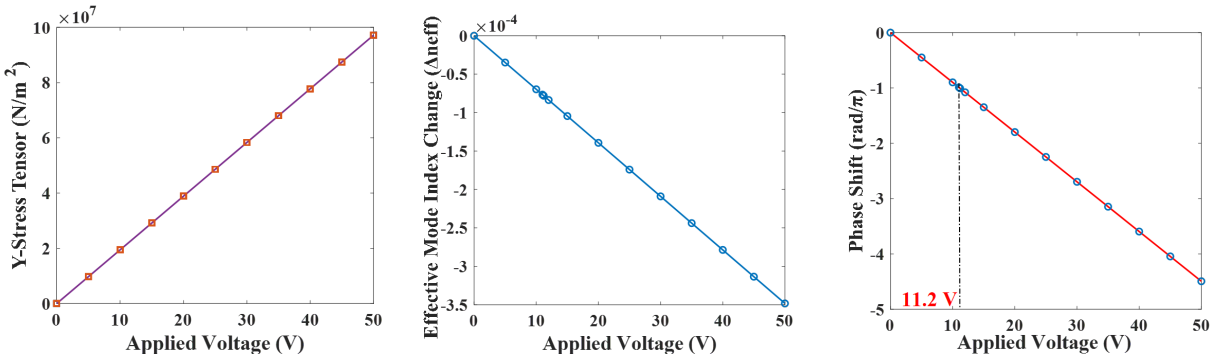



**Figure 4.8:** Simulated y-stress tensor on the SiN waveguide with a 3 µm top cladding and a 3 µm separation between the PZT structures in Design 3. The blue shade represents negative stress, and the red shade represents positive stress.

In Design 3, the positive stress (red shade) is more intense on the SiN core waveguide at 40 V applied voltage with a y-stress tensor of $7 \times 10^7\,\mathrm{N/m^2}$ (Fig. 4.9(a)). The effective index change decreases as the positive stress increases in the waveguide due to the increasing applied voltage on the actuators, as shown in Fig. 4.9(b). In comparison to the other two designs, increased (positive) tensile stress reduces the effective index change in the waveguide structure. The $\pi$-shift is achieved at half wave voltage ($V_\pi$) of 11.2 V (Fig. 4.9(c)).




(a) (b) (c)

**Figure 4.9:** (a) Y-stress tensor in the SiN waveguide and (b) effective index change as functions of applied voltage on the double PZT structures of Design 3. (c) corresponding phase shift as a function of applied voltage for Design 3.

### 4.3.3 Core Waveguide Width Optimization

To further reduce $V_\pi$, the geometry of Design 3 is modified by simultaneously varying the top $SiO_2$ cladding thickness, the distance between the PZT structures, and the width of the SiN core waveguide. The top cladding varies from 3 µm to 2 µm, while the distance between the two PZTs varies from 3 µm to 2 µm. The width of the core waveguide is varied from 2.8 µm to 3.7 µm while maintaining single TE mode confinement to achieve maximum phase shift at low voltage. Fig. 4.10 shows the resulting stress distribution for the optimized PZT actuator-based SiN modulator design with a 2 µm top cladding, a 2 µm separation between two PZT structures, and a 3.7 µm width of the SiN waveguide. This design is referred to as 'Design 4'.

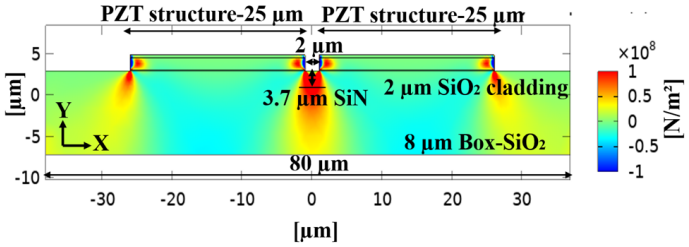
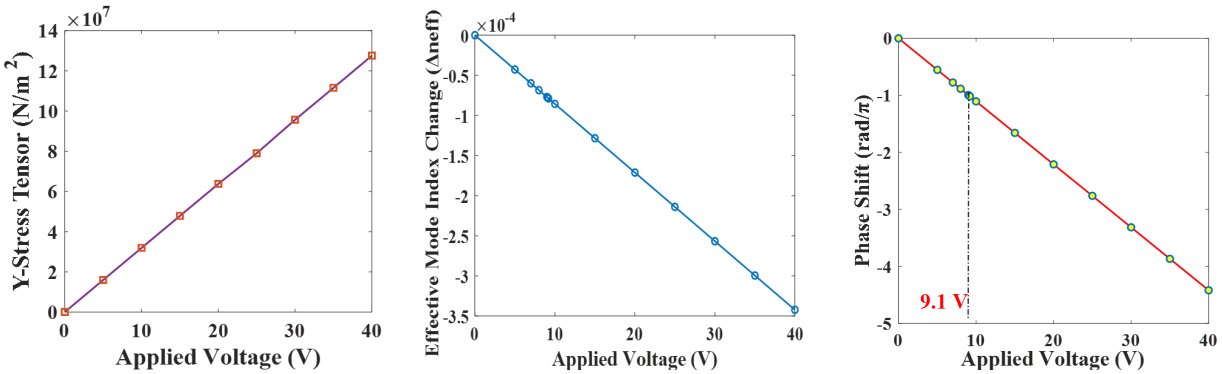



**Figure 4.10:** Simulated y-component of the stress tensor on the SiN waveguide with a 2 µm cladding and a 2 µm distance between the PZT structures in Design 4. The blue shade represents negative stress and the red shade represents positive stress.

Fig. 4.10 illustrates the optimized simulated stress distribution in Design 4, where the PZT structures are spaced 2 µm apart. The red shade ( positive stress) is more intense towards the SiN core waveguide. This causes higher positive stress on the SiN core waveguide.




(a) (b) (c)

**Figure 4.11:** (a) Y-stress tensor in the SiN waveguide and (b)effective index change as functions of applied voltage on the double PZT structures of Design 4. (c) corresponding phase shift as a function of applied voltage for Design 4. Design 4 consists of 2 µm thick $SiO_2$ top cladding, 2 µm distance between two PZT actuators, and 3.7 µm width of SiN core waveguide.

The y-component stress tensor obtained is $1.2 \times 10^8\,\mathrm{N/m^2}$ at 40 V applied voltage. This increased positive stress changes the refractive index changes of the waveguide's material using the stress optic effect, resulting in a reduction of the effective index change as shown in Fig. 4.11(a,b). The $\pi$-shift obtained ($V_\pi$) is at 9.1 V (Fig. 4.11(c)).

## 4.4 Summary of the Optimization Results

This chapter presented the optimization results of a double PZT actuator-based SiN phase modulator with the objective to reduce the half-wave voltage $V_\pi$. The single PZT actuator-based SiN phase modulator achieves a $V_\pi$ of 29 V, with a configuration similar to Lionix's patent design. The design leads to absorption losses since the metal electrode is directly above the waveguide. To reduce this absorption loss and achieve a lower voltage ($V_\pi$), an innovative double PZT actuators-based SiN phase modulator leads to less absorption as the electrodes are not directly above the SiN waveguide. By varying key geometric parameters, such as the distance between PZT actuators, the core waveguide width, and the cladding thickness, the half-wave voltage ($V_\pi$) was reduced. These results are compiled in Table 4.1. These design modifications led to a $\pi$-shift of 9.1 V and reduced the design's footprint, leading to an optimized PZT actuator-based SiN modulator with an enhanced rotational sensitivity for optical gyroscope applications.

**Table 4.1:** Performance comparison between PZT actuator-based SiN modulators

| **Design Number** | **PZT separation (µm)** | **Top cladding thickness (µm)** | **Waveguide width (µm)** | **V$\pi$ (V/cm)** |
|---|---|---|---|---|
| Design Single | 0 | 8 | 2.8 | 29 |
| Design 1 | 22.8 | 8 | 2.8 | 65 |
| Design 2 | 10 | 8 | 2.8 | 33 |
| Design 3 | 3 | 3 | 2.8 | 11.2 |
| Design 4 | 2 | 2 | 3.7 | 9.1 |

# Chapter 5

# Conclusion

This chapter summarizes the thesis content, including the demonstrated results with possible applications. Future works are also discussed.

## 5.1 Thesis Summary

This thesis explored nonlinear and linear phenomena in integrated silicon nitride (SiN) photonic devices to improve the performance of global navigation satellite system (GNSS) architecture using positioning, navigation, and timekeeping (PNT) solutions. SiN is a versatile technology platform with low propagation loss, moderate nonlinear coefficient, low-temperature sensitivity, broadband transparency, and negligible two-photon absorption coefficient. Through those attributes, SiN expands photonic integrated device development for LIDAR, optical clocks, and optical gyroscope sensors. This thesis first introduced SiN

as a nonlinear photonic technology for generating an optical frequency comb (OFC) in a normal dispersion region. It includes the design, simulation, and experimental characterization of a SiN racetrack microring resonator. The characterized SiN microring is used in an analytical simulation to obtain the OFC spectrum. In the second part of the thesis, SiN as an effective photonic integrated technology platform is used in designing a double lead zirconate titanate (PZT) actuator-based SiN phase modulator. Through simulation, its performance is optimized to efficiently provide a phase shift from a relatively low voltage. A summary of each chapter is presented below.

Chapter 2 described the recent advancements related to SiN microring resonators for OFC generation with the fundamental principles, specifically, four-wave mixing (FWM), dispersion, and the nonlinearity. It also discussed the recent development related to PZT-based SiN phase modulators, the fundamental principles of optical gyroscope, stress-optic and piezoelectric effect leading to phase change. Chapter 3 outlines the design, simulation, and experimental characterization of SiN microring racetrack resonators to achieve an acceptable normal dispersion -359 ps/nm.km at 1547 nm. Experimental characterization of the SiN racetrack microring resulted in a loaded quality factor ($Q$) of 77,800, acceptable for OFC generation. Through simulation, the demonstrated resonator can generate six comb lines through non-degenerate four-wave mixing (FWM). Chapter 4 details the investigation of an efficient double PZT actuator-based SiN phase modulator. By varying key geometric parameters, the reduced half-wave voltage ($V_\pi$) was 9.1 V from

65 V, improving the sensitivity of optical gyroscopes.

## 5.2 Future Works

The work presented in this master's thesis can be pursued by first completing the experimental validations. The results may open the door to different possible research directions.

**1) OFC Generation Experimental Validation**

The fabricated SiN die for this research work exhibits a minimum fiber-to-chip coupling loss at 1547.19 nm of approximately 9.5 dB, leading to a total coupling loss of 19 dB, much higher than the expected 11 dB from the process design kit (PDK). The die included edge coupling as well, consisting of an inverse taper with a waveguide width of 1000 nm at the edge of the chip, enabling the mode conversion from the optical fiber to the SiN waveguide. For the optical fiber probe, I used a tapered fiber tip with index-matching liquid that brought the optical fiber diameter of 8 µm down to 2.5 µm. However, the alignment was challenging, and I was unable to obtain a coupling loss better than 16 dB per facet due to a lack of time to acquire the necessary experience. Our research team had experimentally performed edge coupling. Graduate students measured less than 5 dB loss per edge coupling facet [157] as well as less than 4 dB per face in another project where the coupling was to a polymer-based die [158]. With more time dedicated to the experimental testbed and the necessary adjustment needed to the optical table to minimize fluctuation, a total coupling loss of at

most 5 dB (as indicated by ANT for the tapered edge coupling designs) could be obtained.

The proposed experimental setup for OFC generation using the designed SiN microring racetrack resonator is illustrated in Fig. 5.1. The setup employs two tunable lasers operating at 1546.56 nm and 1547.02 nm wavelengths, respectively, each with an optical power of 1.2 dBm. The recommendation is to use an acquired high-power low-noise erbium-doped fiber amplifier (EDFA) to compensate for the device and experimental testbed losses. The combined laser signal is amplified with a gain of 27.3 dB. The optical signal should have a total power of 24.3 dBm. The edge coupler expected loss is 5 dB/facet for a 2.5 µm mode field diameter (MFD). On-chip, the optical power of each pump is needed to be 17.8 dBm (60 mW), sufficient to generate OFC in the microring and to generate four comb signals at 1547.52 nm, 1546.08 nm, 1545.60 nm, and 1548 nm. The resultant OFC spectrum can be measured by an optical spectrum analyzer (OSA).

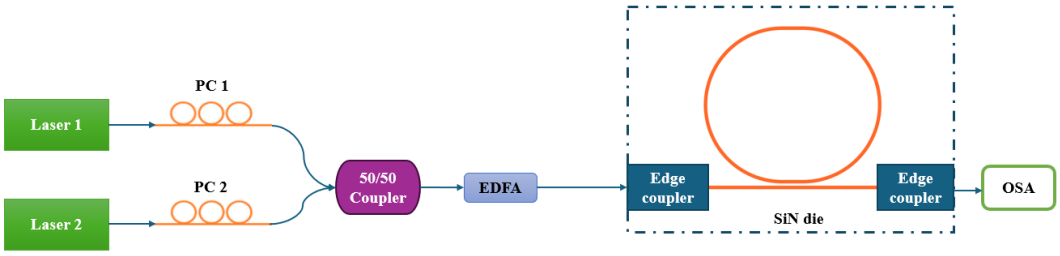



**Figure 5.1:** Proposed an experiment testbed for OFC generation in the SiN microring racetrack resonator. EDFA: Erbium-doped fiber amplifier (Pritel, LNHPFA-33-IO-NMA), PC: polarization controller, OSA: optical spectrum analyzer.

### 2) PZT Actuator-based SiN Modulator Experimental Validation

The double PZT actuator-based SiN phase modulator optimized in this thesis can be fabricated following similar work reported by LioniX International in the Netherlands [132]. The TriPleX™ fabrication process developed by LioniX is reproduced in Fig. 5.2 and features an asymmetric double stripe for achieving low propagation loss of $< 0.1$ dB/cm. The SiN core consists of two layers: a 75 nm bottom layer and a 175 nm top layer, separated by a layer of silica ($SiO_2$). The waveguide has a default width of 1.1 µm, with 3 µm of $SiO_2$ top cladding above it, separating the waveguide from the PZT structure. The PZT layer (green) is 1.5 µm thick and includes a 100 nm Pt bottom electrode (orange), serving as a common ground for all devices on the wafer. A top electrode (orange) is applied locally above the PZT to create an electric field between the top and bottom electrodes. The TriPleX™-based PZT modulator experimentally demonstrated a phase shift of $\pi$ at an applied voltage of 16 V for a device length of 1 cm.

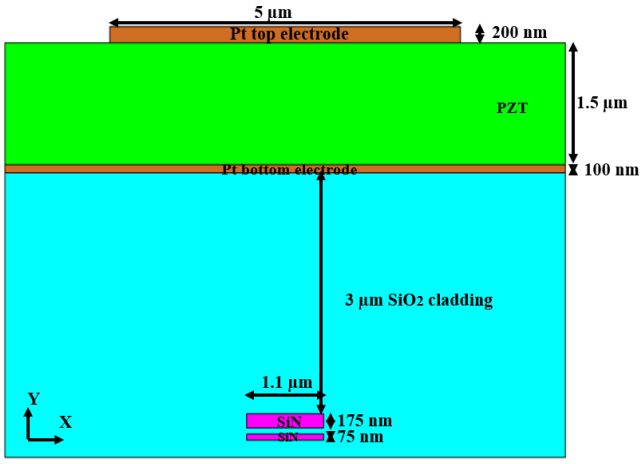



**Figure 5.2:** The cross-sectional waveguide with the PZT actuation featuring an asymmetric double stripe (ADS) TriPleX™ silicon nitride waveguide.

The performance of the optimized double PZT actuator-based SiN phase modulator can be experimentally validated by designing a 1550 nm SiN-based Mach-Zehnder Interferometer (MZI) with one arm incorporating the PZT actuators featuring a 1 cm electrode length as in [132]. The double PZT actuator in one arm senses stress and leads to a phase shift through the stress-optic effect, while the other arm serves as a reference path.

To characterize the phase shift, a tunable continuous-wave (CW) laser can be injected into the input port of the MZI. A DC voltage of 9.1 V can then be applied to the top electrodes of the double PZT actuators, while the bottom electrodes are grounded (0 V) using DC voltage probes. An optical power meter is connected to the output port of the MZI to monitor the output light. As the phase shift varies with the voltage applied to the PZT actuator, the interference pattern changes, leading to a change in the intensity of the output signal. Lionix investigated the impact of top cladding thickness on the performance of their PZT-based SiN actuators by varying the cladding from 3 µm to 2 µm. For a 15 mm actuator length, they reported half-wave voltages ($V_\pi$) of 12 V for the 3 µm cladding and 11 V for the 2 µm cladding [159]. In our simulations of optimized double PZT actuators on SiN waveguides, we observed $V_\pi$ values of 11.2 V for a 3 µm top cladding and 9.1 V for a 2 µm top cladding. These results suggest that employing Lionix's fabrication process for our actuator designs should not adversely affect the optimized $V_\pi$ performance.

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